

AOTEMPO-MEDIATED CROSSLINKING OF POLYPROPYLENE WITH MIXED INITIATOR SYSTEMS

by

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Abstract

4-Acryloyloxy-2,2,6,6-tetramethylpiperidine-N-oxyl (AOTEMPO) can be used to modulate the onset, rate, and yield of radical-mediated crosslinking of polypropylene (PP) with peroxide initiators. The possibility of using mixed peroxide formulations comprised of dicumyl peroxide (DCP) and 1,1-bis(tert-butylperoxy)-3,3,5-trimethylcyclohexane (L-231) to adjust the induction time of AOTEMPO cures was investigated as a means of providing flexibility to formulation developers.

In addition, pendant acrylate groups introduced by AOTEMPO trapping of alkyl macroradicals have been exploited to support a secondary UV-curing process. This involves a two-step sequence of a conventional peroxide-initiated alkoxyamine formation, followed by photocuring of polymer-bound acrylate functionality. The efficacy of this alternative approach to PP crosslinking has been demonstrated using a range of type I photo-initiators.

Co-Authorship

This research was conducted independently by the author under the supervision of Dr. J. Scott Parent.

Disclosure of sources of Funding and Conflicts of Interest

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Statement of Originality and Disclosure of Use of Artificial Intelligence

In the preparation of this thesis, artificial intelligence was not used. No AI-generated content, figures, or citations were used in this thesis.

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List of Abbreviations

PP – Polypropylene

iPP – Isotactic Polypropylene

sPP – Syndiotactic Polypropylene

PE – Polyethylene

DI – Dispersity Index

AOTEMPO - 4-acryloyloxy-2,2,6,6,-tetramethylpiperidine-1-oxyl

DCP – Dicumyl Peroxide

t_{ind} – Induction time

L-231 - 1,1-Bis(tert-butylperoxy)-3,3,5-trimethylcyclohexane

H-atom – Hydrogen atom

AE – Abstraction efficiency

β -scission – Beta scission

TR – Trapping ratio

PF – Peroxide Fraction

ASTM – American Society for Testing and Materials

APA – Advanced Polymer Analyzer

G' – Dynamic storage modulus

wt% - Weight percent

$\mu\text{mol/g}$ – micromole per gram polymer

UV – Ultraviolet

DCM – Dichloromethane

PI – Photo-initiator

I2959 - 2-Hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone

L-TPO - Diphenyl(2,4,6-trimethylbenzoyl)phosphine oxide

I819 - Phenylbis(2,4,6-trimethylbenzoyl)phosphine oxide

M_n – Number average molecular weight

M_w - Weight average molecular weight

MFI – Melt Flow Index

SE – Standard Error

PP-67k – Polypropylene (M_n = 67,000 g/mol)

PP-97k – Polypropylene (M_n = 97,000 g/mol)

Chapter 1

Introduction

Polypropylene (PP) is one of the most widely used thermoplastic polymers, valued for its exceptional balance of mechanical properties, chemical resistance, processability, and cost-effectiveness. The versatility of its properties can be further enhanced through copolymerization, blending, and functionalization, enabling tailored performance for specialized applications such as membrane technology, composite manufacture and fibre reinforcement [1]. Polypropylene can be produced in the three stereoisomeric forms shown in Figure 1. The stereogenic centres in isotactic PP (iPP) have the same chirality, which leads to relatively high degree of crystallinity and a melting point around 171°C. In syndiotactic PP (sPP with $\approx 30\%$ crystallinity) the chirality of tertiary main chain carbons is alternating, while atactic PP has a random chirality that leads to an amorphous morphology with no defined melting point [2].

The linear chain structure of most commercial grades of iPP makes it ill-suited to applications that require melt elasticity and high extensional viscosity. Reactive melt processing methods, such as introducing long-chain branching or crosslinking, are used to enhance the polymer framework. Conventional techniques involve peroxide-induced radical reactions and high energy irradiation methods [3]. Extensive crosslinking of PP introduces a three-dimensional network of covalent bonds, which can enhance thermal and mechanical stability of the material in the solid state [4].

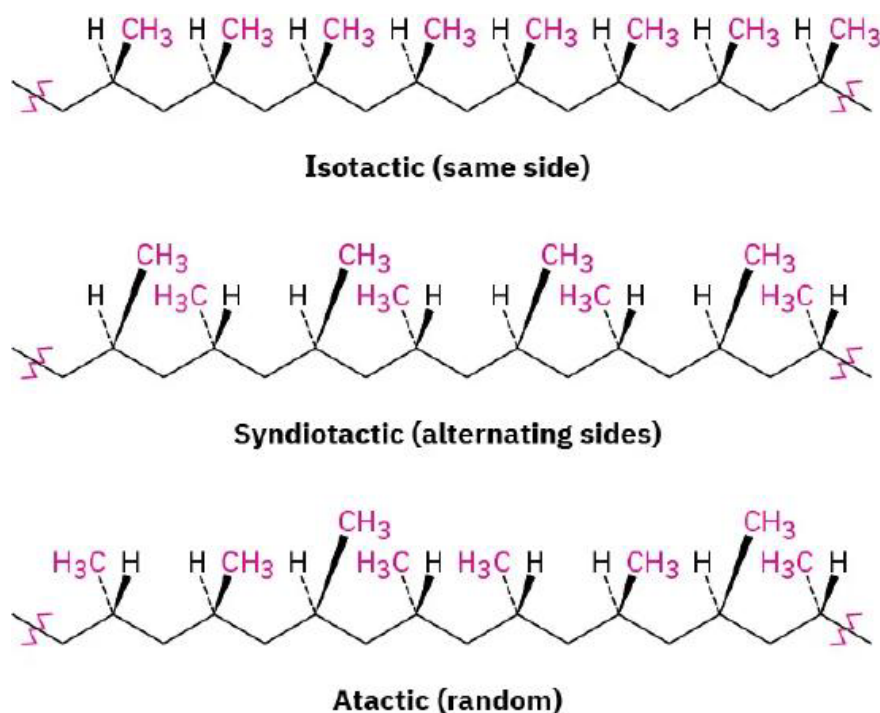


Figure 1: Stereospecific forms of PP

[5]

1.1. Cure reversion in polypropylene

A major problem with conventional peroxide-initiated chemistry is the deterioration in mechanical properties of PP [6]. Unlike polyethylene (PE), which crosslinks readily through peroxide-initiated radical chemistry, polypropylene undergoes competitive radical-induced degradation reactions that reduce molecular weight and compromise mechanical properties. A greater extent of crosslinking in polypropylene can be achieved at higher peroxide concentrations, but the process is complicated due to simultaneous chain scission and other side reactions like oxidation, which may introduce carbonyl and unsaturated groups into the polymer network [4].

During polyolefin crosslinking, peroxide initiators generate alkoxy radicals that abstract H-atom from the polymer backbone. In the case of PP (Fig. 2), the preference is for tertiary alkyl macroradical formation, whose susceptibility to β -scission can yield a vinylidene group

and a secondary alkyl macroradical [7]. Since the probability of H-atom donation by a polymer chain is proportional to its number average molecular weight, the largest chains in a sample of PP are more susceptible to cleavage than the smallest chains. As a result, treatment of PP with peroxide leads to reductions in molar mass and Dispersity Index (DI). This is called controlled degradation or peroxide shifting in the polymer industry [8].

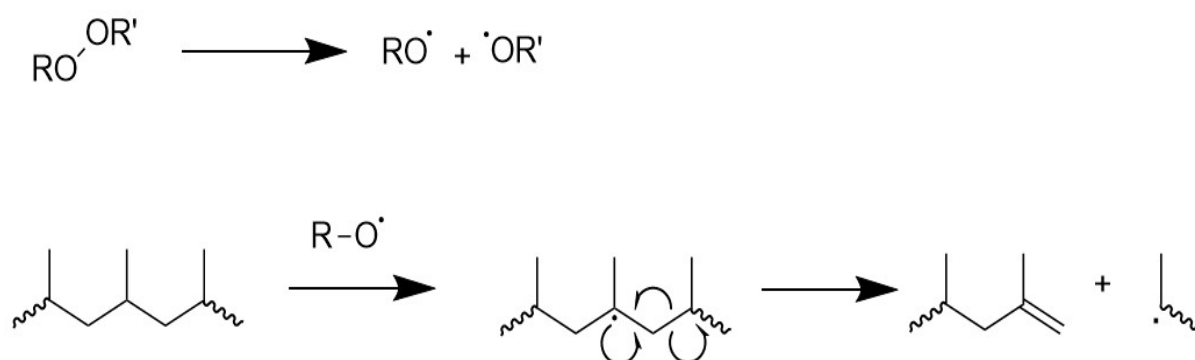


Figure 2: Peroxide-initiated H-atom abstraction and subsequent β -scission reaction of polypropylene.

Note that this PP degradation pathway competes with radical termination reactions. As such, chain scission and degradation dominate the reaction outcome at peroxide concentrations of practical concern, rendering peroxide only-crosslinking of PP highly ineffective. The net result can be quantified by measurement of the decline in dynamic storage modulus (G') over time, due to the sensitivity of G' to chain length and architecture [9].

Reaction temperature strongly affects the balance between crosslinking and chain scission during PP modification. Since β -scission requires higher activation energy than bimolecular radical termination, lower temperatures generally favour radical combination [4]. However, the high melting point of iPP imposes a lower limit ($T > 165\text{ }^{\circ}\text{C}$) on polymer processing

temperature. Hence, an alternate approach is required to overcome the challenge of PP degradation.

1.2. Hydrogen-atom abstraction efficiency (AE)

The extent of H-atom abstraction from a polymer by a peroxide is often quantified in terms of H-atom abstraction efficiency (AE). Dicumyl peroxide (DCP) is a widely used peroxide initiator for polyolefin functionalisation due to its commercial availability and well characterised decomposition behaviour [10]. The decomposition of DCP to alkoxy radicals leads either to cumyl alcohol by H-atom abstraction or to methyl radical + acetophenone by cleavage (Fig. 3). Hence the susceptibility of a polyolefin to H-atom abstraction by DCP can be quantified by the relative quantities of cumyl alcohol and acetophenone [11]. A general expression for AE can be formulated based on byproduct concentrations, as shown below.

$$AE = \frac{[\text{cumyl alcohol}]}{[\text{cumyl alcohol}] + [\text{acetophenone}]} \quad (1)$$

The study by Garrett et al. [12] determined an AE of 0.37 for PP and 0.56 for PE at 160°C for a peroxide-only formulation containing 7.2 µmol DCP/g PP. This means that only 37% of the radicals generated by DCP decomposition succeed in abstracting a hydrogen atom from PP chains to create polymer macroradicals. The lower abstraction efficiency in polypropylene results from entropic control where steric hindrance from methyl branches creates energy barriers that exceed the thermodynamic advantages of weaker tertiary C-H bonds, leading to suppressed reactivity at secondary positions [13].

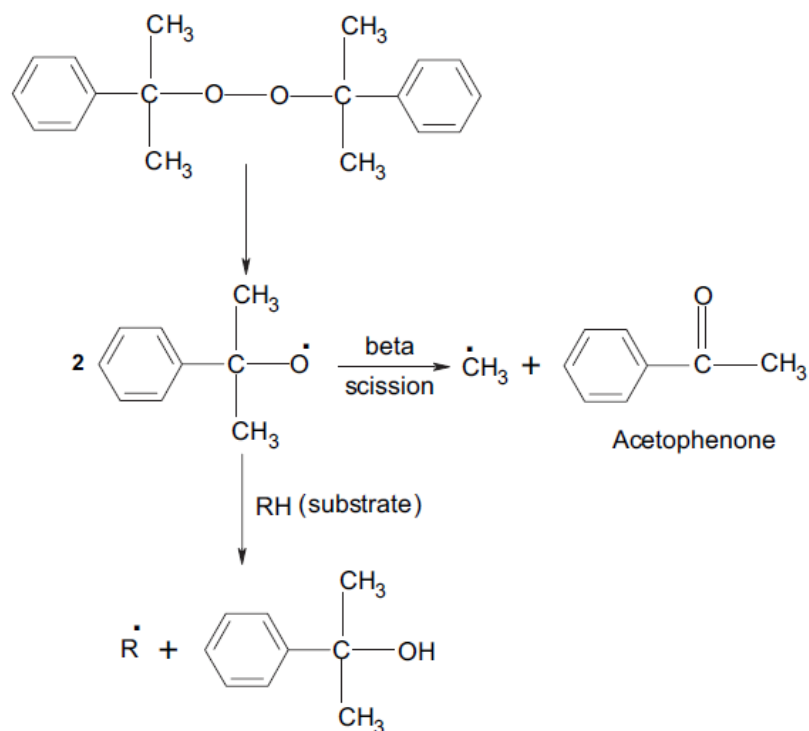


Figure 3: Decomposition pathway of DCP

[14]

1.3. Quantifying cure yield

The extent of crosslinking is generally quantified by two measurement techniques. Gel contents for PP are determined by the weight fraction of material this is insoluble in boiling xylenes. This gel residue possesses a branch-on-branch chain architecture whose molecular weight is so large as to render it insoluble in a good thermodynamic solvent [15]. A more precise measure of crosslink density centres on the dynamic storage modulus (G'), which is a measure of the elastic response of a polymer melt to an applied deformation. This is determined by dynamic mechanical analysis (DMA) or oscillatory rheology on a melted PP sample. Introducing long chain branching or a fully gelled chain architecture to iPP affects G' through the action of chain entanglement, which increase relaxation times of chains in response to an oscillatory strain. As such, the ultimate storage modulus ($G'_{\max}$) can be used

to estimate the degree of crosslinking, though it is also affected by factors such as the molecular weight and density of the polymer [16, 17]. Gel content quantifies the fraction of crosslinked material, and storage modulus provides a sensitive measure of mechanical rigidity imparted by crosslinks. Using both allows for a robust assessment of crosslink density in polymer systems.

1.4. Nitroxyl-mediated crosslinking

Nitroxide-mediated polymerization (NMP) (“aminoxyl-mediated radical polymerization” or AMRP) [18] is a form of controlled radical polymerization (CRP) that employs nitroxyl radicals for reversible deactivation of propagating radicals. The principal mechanism is the formation of an adduct or alkoxyamine (dormant species) by the reversible combination of a polymer macroradical with a nitroxyl. A polymer radical formed using a suitable initiator is reactive not only to the nitroxyl but also to other monomers and polymer radicals (forming a macroradical). Since the nitroxyl is only reactive toward carbon centred radicals, it does not interfere with the formation of polymer macroradical [19]. In the case of controlled radical polymerization, the alkoxyamine undergoes repeated reversible homolytic cleavage of the C-O bond of the alkoxyamine to provide propagating radicals in a pseudo-living condition. Using NMP, it is possible to attain very narrow mass distribution and chain ends that can be re-activated. So, it is possible to synthesize polymers with narrow dispersity and controlled chain architecture [20].

1.5. TEMPO and its derivatives

Nitroxyls have been successfully used as macroradical inhibitors due to their distinctive molecular space structure and single electron delocalization [21]. The stability of nitroxide radicals is attributed to steric hinderance of bulky groups that prevent coupling to the N-

centred resonance form. A well-studied example is (2,2,6,6-tetramethylpiperidine-1-yl)oxyl or TEMPO and its many derivatives which are cyclic nitroxides that possess a long shelf life. The persistence of these nitroxyl radicals is primarily attributed to the delocalization of the unpaired electron across the N–O bond, and steric stabilization provided by the four α -methyl substituents adjacent to the N–O center [22]. Certain derivatives of TEMPO bear pendant functionality (Fig. 4) which can be utilized for polymer modification.

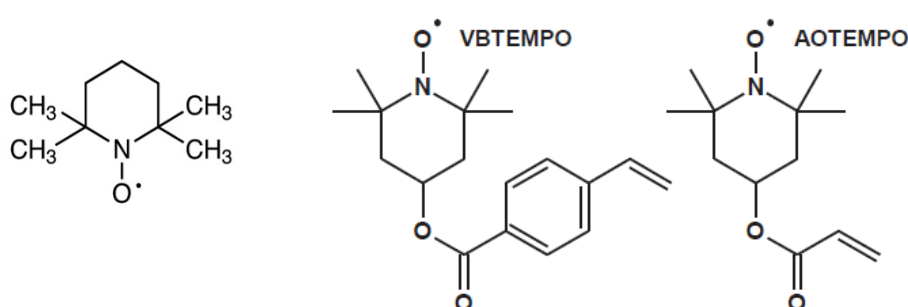


Figure 4: TEMPO and its functional derivatives

1.6. Crosslinking and Scorch protection

The conventional method of crosslinking polyolefins such as polyethylene involves the thermolysis of organic peroxides to generate polymer radicals in a melt state. Scorch, or premature crosslinking, arises from insufficient mixing and rapid peroxide decomposition, resulting in localized regions of excessive crosslinking and chain scission within the polymer melt. This compromises processability, mechanical properties, and leads to surface defects [23]. The irreversible trapping of macroradicals by nitroxyl can provide useful scorch protection. Chaudhary *et al.* [24] explored the use of 4-hydroxy TEMPO (HO-TEMPO) for scorch protection in the crosslinking of two grades of low-density polyethylene (LDPE) at 140°C. It was noted that, in contrast to previously utilised scorch retardants like hindered

phenols, HO-TEMPO provided a flat induction time at the beginning. This reflected the rapid trapping of polymeric alkyl radicals derived from peroxide decomposition in the first few minutes. However, there was an overall loss in degree of crosslinking due to quenching of the macroradicals that are needed to produce crosslinks. Using another TEMPO derivative bis-TEMPO, scorch suppression was still achieved to some extent, but grafting of its pendant nitroxyl groups onto other polymer chains led to an increase in the final degree of crosslinking. This showed the potential of TEMPO and its derivatives with pendant functional groups to not only suppress scorch by introducing an induction period (Fig. 5), but also to enhance the extent of crosslinking.

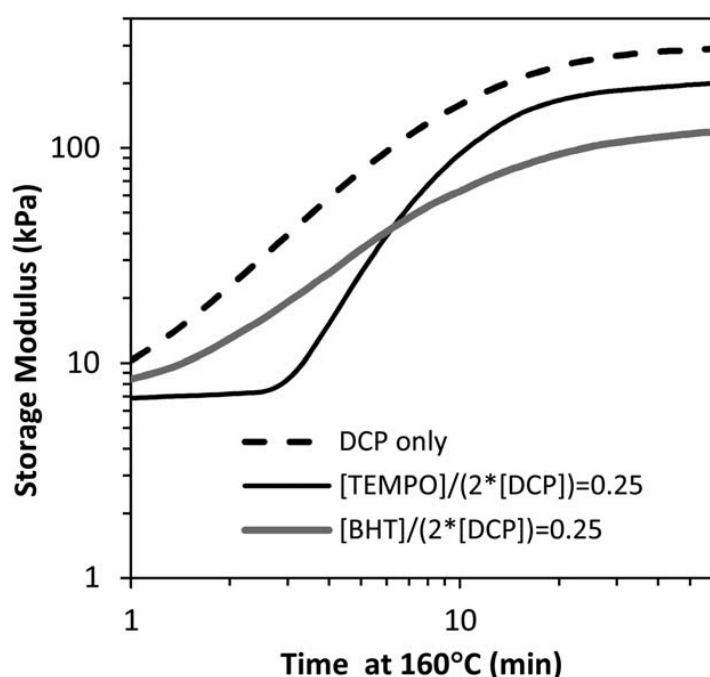


Figure 5: Difference in dynamics of LLDPE cure with TEMPO and a hindered phenol like BHT [25]

Hyslop *et al.* [26], studied the potential of 4-acryloyloxy-2,2,6,6-tetramethylpiperidine-N-oxyl (AOTEMPO) in trapping macroradicals to create macromonomers (alkoxyamines) with pendant acrylate functionality. The trapping of radicals created the desired induction time, which was followed by oligomerization of acrylate functionality to provide the desired thermoset. The initial work focused on linear low-density polyethylene (LLDPE), but a general expression for the induction time (t_{ind}) is time was determined as follow:

$$t_{ind} = -\frac{1}{k_d} \ln \left[1 - \frac{[\text{nitroxyl}]}{2[\text{ROOR}]} \right] \quad (2)$$

where k_d is the first-order rate constant for initiator decomposition at the reaction temperature. The term $[\text{nitroxyl}]/2[\text{ROOR}]$ represents the ratio of nitroxyl to peroxide-derived radicals, defined as trapping ratio (TR). The TR represents the fraction of initiator-driven radicals quenched by the nitroxyl. The expression for t_{ind} was derived under the assumption that induction time ends when all nitroxyl functionality is consumed, at which point the peroxide conversion is equal to TR. As such, the induction period is governed by the TR and the initiator half-life.

$$\text{TR} = \frac{[\text{nitroxyl}]}{\text{radicals per peroxide molecule} \times [\text{peroxide}]} \quad (3)$$

A more recent publication by McLaren *et al.* [27] focused on extending the use of AOTEMPO to functionalise polypropylene (Fig.6). The initial step (Phase 1) of radical trapping is instantaneous so that macroradical lifetime is significantly reduced, effectively eliminating the concern of PP degradation. The portion of macroradicals converted to alkoxyamine depends on the TR and the H-atom abstraction efficiency. In the second phase, alkoxyamines act as monomer units whose oligomerisation of C=C functionality helps to recover the loss in crosslinking due to the initial quenching. Since only a fraction of the generated



[27]

1.7. Research Objectives

This report summarizes investigations of AOTEMPO-mediated cures of PP through different initiator systems. In Chapter 2, the use of a mixed initiator system of two peroxides to control cure onset and kinetics is described. The primary objectives are to evaluate the change in induction period and the extent of crosslinking obtained by utilising different combinations of the peroxide mix.

In Chapter 3, an alternate two-step cure process is described. The initial functionalisation of PP is carried out using a peroxide initiator at a high trapping ratio, while the crosslink formation using the C=C functionality is achieved via UV cure using a suitable photo-initiator. The objective of this chapter is to minimise the effects of β -scission to the maximum extent by using a trapping ratio of 0.99 such that nearly all macroradicals are effectively quenched by AOTEMPO. The subsequent UV crosslinking is achieved at room temperature and hence thermal degradation during crosslinking is avoided.

Chapter 2

Mixed Initiator Systems for Controlling Cure Dynamics

While AOTEMPO formulations can address the fundamental cure reversion problem, controlling the precise timing and dynamics of the crosslinking process requires careful manipulation of radical generation rates throughout the cure cycle. This gives the inspiration for the use of a dual initiator system. Typically, a low-temperature peroxide is selected to ensure rapid initiation, while a high-temperature peroxide decomposes comparatively slower but drives the reaction to completion [29]. This dual-initiator strategy exploits the difference in decomposition kinetics of peroxides.

Pojman *et al.* [30] explored the use of a dual initiator system of benzoyl peroxide (BPO) and t-butyl peroxide (tBPO) for frontal polymerization of methacrylic acid. In frontal polymerization a self-sustaining reaction front propagates through the monomer driven by the heat released from polymerization. The dual-initiator system exploits the difference in thermal stability of BPO and tBPO. BPO decomposes first to initiate the reaction and generate heat, which raises the local temperature sufficiently to trigger decomposition of tBPO. BPO by itself is highly reactive but produces rapid reaction front with poor conversion due to a high instantaneous phenyl radical concentration which terminates growing polymer chains. tBPO is a more stable initiator that provides better conversion at a slower rate. The combination of tBPO and BPO resulted in a higher front velocity compared to tBPO alone, giving rapid propagation and good conversion. This combination also provided a sharp but stable temperature profile, and the high ignition temperature enabled more thermal diffusion.

Another study on polyurethane-based thermosets discussed the use of a low temperature initiator like bis[4-*t*-butylcyclohexyl]-peroxy dicarbonate (TBIC), along with high temperature initiators to initiate an exothermic crosslinking reaction. The addition of TBIC along with *t*-butylperoxybenzoate (TBPB) lowered the onset and peak temperature indicating faster cure initiation. These effects are attributed to TBIC's ability to rapidly generate radicals at lower temperatures. Optimizing the combination of peroxide initiators is critical for attaining a high level of cure [31].

2.1. DCP and L-231

A similar approach can be adopted in the crosslinking of polypropylene using a comparatively stable initiator like DCP with a highly reactive initiator like Luperox 231 to create a synergistic peroxide initiator system.

Luperox 231 or 1,1-bis(*tert*-butylperoxy)-3,3,5-trimethylcyclohexane is a conventional peroxide initiator utilized in radical polymerization due to its high solubility in organic media. It is a room-temperature stable liquid that is readily dissolved in organic solvents like acetone [32]. A kinetic model by Dan *et al.* [33] predicted the half-life of L-231 at 0.9 mins for 160°C and 0.3 at 170°C with an activation energy of 128 kJ/mol. At the processing temperature for polypropylene, which is about 165°C, L-231 is expected to be consumed early in the thermal cycle, providing an initial burst of radicals. This helps to provide radicals instantaneously and initiate the AOTEMPO trapping phase.

system is utilised as opposed to single initiator system (TBPB alone). This can be attributed to accelerated inhibitor consumption resulting from enhanced radical flux due to MEKP and its synergistic effect on TBPB.

Similarly, according to the ratio of L-231 used in conjunction with DCP, there should be a decrease in induction time. At the processing temperature for PP (around 165°C) the half-life of L-231 is lower compared to DCP. Hence, L-231 exhibits faster radical generation kinetics compared to DCP, generating radicals at a quicker rate which are immediately scavenged by a portion of the AOTEMPO. This leads to faster depletion of the inhibitor pool (AOTEMPO) to a certain extent, resulting in a reduction in overall induction time. So, the introduction of small quantities of L-231 to replace DCP can lower the induction time when required. The equation for t_{ind} can be adjusted to find the induction time for the combination of L-231 and DCP.

2.3. Industrial Significance

The use of a mixed-initiator system enables precise matching of processing requirements with cure dynamics. This allows processors to optimize cycle times and final properties independently. For example, a higher proportion of fast initiator would shorten the induction time, but it might affect the final crosslink density.

Reducing induction time enables faster production cycles and higher throughput in industrial processes. This is particularly beneficial in applications such as extrusion coating, cable insulation, and automotive component moulding, where faster curing shortens processing time while still achieving the required crosslink density and final mechanical performance.

2.4. Experimental

2.4.1. Materials

Isotactic polypropylene (PP, $M_w=340,000$ g/mol, $M_n=97,000$ g/mol, MFI= 4g/10min), dicumyl peroxide (DCP, 98%), acetone and Luperox 231 were used as received from Sigma-Aldrich. 4-Acryloyloxy-2,2,6,6-tetramethylpiperidine-1-oxyl (AOTEMPO) was provided by Ingenia Polymers.

2.4.2. Sample PP formulation

PP pellets were ground to a powder using a tabletop grinding machine. The powdered PP (5g) was divided evenly in glass jars. Standard solutions of the desired reagents DCP, L-231 and AOTEMPO in acetone were prepared according to the required weight percents. The powdered PP was coated with these solutions and mixed internally on a jar mill roller for 15 mins and subsequently dried by exposing to a stream of air.

The amount of DCP used was 2 wt% of the sample PP. The quantity of radicals created by DCP is represented as:

$$\frac{2\text{g DCP}}{100\text{ g PP}} \times 5\text{gPP} \times \frac{1}{270.37 \frac{\text{g}}{\text{mol}}} \times \left(6.022 \times 10^{23} \frac{\text{molecules}}{\text{mol}}\right) \times \frac{2\text{ radicals}}{\text{molecule}} = 4.455 \times 10^{20} \text{ radicals} \quad (4)$$

Therefore, the amount of L-231 used in combination with DCP should be adjusted in such a way that it provides 4.455×10^{20} radicals. Therefore, by reverse calculation,

$$4.455 \times 10^{20} \times \frac{1\text{ molecule}}{2\text{ radicals}} \times \frac{1\text{ moles}}{6.022 \times 10^{23} \text{ molecules}} \times 302.45 \frac{\text{g}}{\text{mol}} \times \frac{1}{5\text{g PP}} = 2.237 \text{ weight\%} \quad (5)$$

Hence, the weight percent of L-231 used was 2.237 wt% in combination with 2 wt% of DCP.

PP formulations with different peroxide ratios were prepared as described in Table 1.

2.4.3. Instrumentation and Analysis

Time-resolved rheological measurements were acquired using an Advanced Polymer Analyzer 2000 (APA, Alpha Technologies), equipped with biconical plates. The tests were conducted at 165 °C, with a frequency of 1 Hz and a strain amplitude corresponding to a 3° arc.

Gel contents of PP samples were measured by cutting into pieces of approximately 0.1 g before wrapping in pre-weighed 120-mesh sieve cloth. The sample was placed in refluxing xylenes stabilized with 100 ppm BHT for 4 hours (ASTM D 2765-01). The sample was retrieved from the solution, and residual material was dried in a vacuum oven at 90°C to a constant weight. The difference in final mass compared to initial mass as a percentage indicates the gel fraction.

2.5. Results

2.5.1. Induction time formulation

The general expression for the induction time provided by an AOTEMPO-mediated crosslinking formulation involving a single initiator has been established in terms of $TR = [\text{nitroxyl}]/2 \cdot [\text{ROOR}]$ and the rate constant for peroxide decomposition, k_d [26]. However, in a dual-initiator system the contribution of both peroxides to the induction time needs to be considered in terms of their respective decomposition rates at the polymer processing temperature.

Since both peroxides decompose according to first-order kinetics, their concentration ($\mu\text{mol/g}$) can be expressed generally as:

$$[\text{ROOR}] = [\text{ROOR}]_0 e^{-k_{\text{ROOR}} t} \quad (6)$$

Where $[\text{ROOR}]_0$ is the initial peroxide loading, while $[\text{ROOR}]$ is the concentration of peroxide at given time t . k_{ROOR} is the first order rate constant for the homolysis of the corresponding peroxide at the reaction temperature.

The total amount of peroxide charged into the PP formulation, considering the dual-initiator system of DCP and L-231:

$$[\text{ROOR}]_0 = [\text{DCP}]_0 + [\text{L} - 231]_0 \quad (7)$$

The total concentration of radicals generated by both peroxides at time t can be expressed as:

$$[\text{Radicals}]_{\text{Total}} = 2 [\text{DCP}]_0 (1 - e^{-k_{\text{DCP}} t}) + 2 [\text{L} - 231]_0 (1 - e^{-k_{\text{L-231}} t}) \quad (8)$$

The relative concentration of L-231 and DCP is defined as the term peroxide fraction (PF):

$$\text{PF} = \frac{[\text{L} - 231]}{[\text{DCP}] + [\text{L} - 231]} \quad (9)$$

The initial concentration of both initiators can be re-written in terms of PF and total peroxide loading $[\text{ROOR}]_0$ as follows:

$$[\text{DCP}]_0 = (1 - \text{PF}) [\text{ROOR}]_0 \quad (10)$$

$$[\text{L} - 231]_0 = \text{PF} [\text{ROOR}]_0 \quad (11)$$

Since, the induction time (t_{ind}) is the point where all nitroxyl radicals are consumed and crosslinking starts, the total concentration of radicals consumed at t_{ind} would be equal to the concentration of nitroxyl charged to the formulation. As such, the equation can for radical concentration can be re-written in terms of PR and $[\text{ROOR}]_0$ as follows:

$$[\text{Radicals}] = [\text{Nitroxyls}] \quad (\text{at } t = t_{\text{ind}})$$

$$[\text{Nitroxyl}] = 2 \text{ PF } [\text{ROOR}]_o (1 - e^{-k_{\text{DCP}} t_{\text{ind}}}) + 2 (1 - \text{PF}) [\text{ROOR}]_o (1 - e^{-k_{\text{L-231}} t_{\text{ind}}}) \quad (12)$$

Since the trapping ratio $\text{TR} = [\text{Nitroxyl}] / 2 [\text{ROOR}]$ the above equation can be modified to develop a relation between the induction time and the trapping ratio of a formulation:

$$\text{TR} = (1 - \text{PF})(1 - e^{-k_{\text{DCP}} t_{\text{ind}}}) + \text{PF} (1 - e^{-k_{\text{L-231}} t_{\text{ind}}}) \quad (13)$$

Four cases of initiator loading combinations have been studied to test this relationship. The general idea is that, owing to its more rapid decomposition kinetics, higher L-231 concentrations should reduce the induction time, with PF values increasing from 0 to 1 resulting in a significant decrease in t_{ind} .

At the 165°C processing temperature used for PP, the half-life of DCP is 3.4 mins and that of L-231 is 0.3 mins. The expression relating TR, PF and t_{ind} can be used to predict the trend of change in t_{ind} with TR at a specific PF. As evident from Fig 8, the increase in PF or amount of L-231 added, clearly leads to a reduction in induction time. The induction time corresponding to $\text{PF}=1$ is very low which might not be favourable for processing before the onset of crosslinking. A PF value of 0.5 provides a considerable induction time, enabling sufficient time for processing or reshaping of the melt. Therefore, PF values of 0.5 and above are considered feasible for practical application.

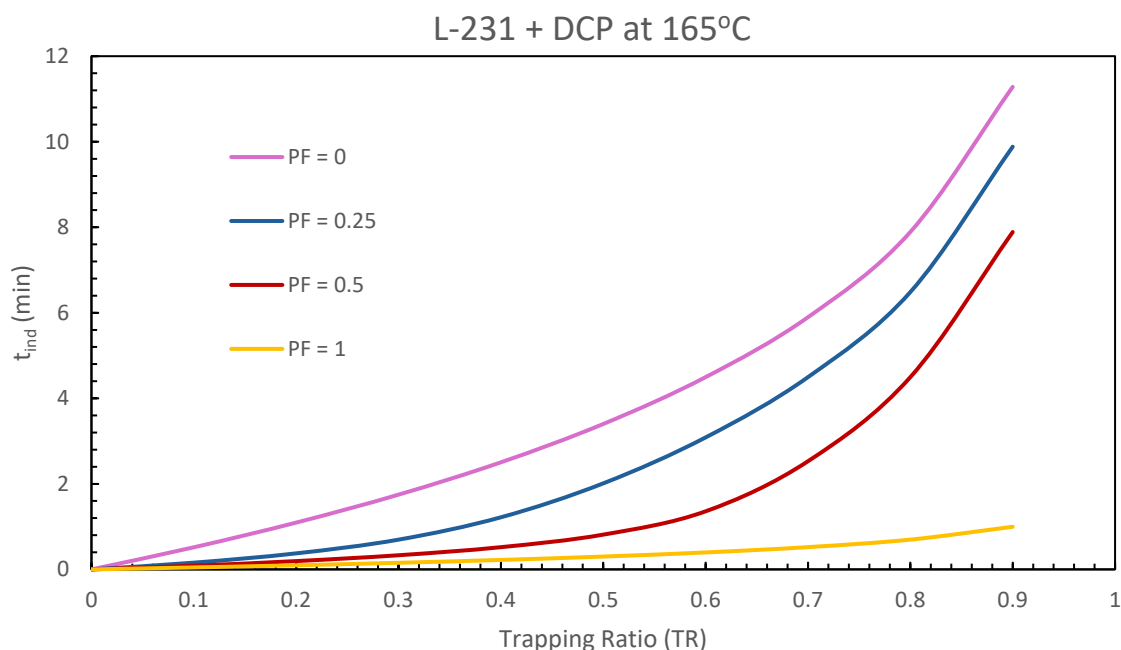


Figure 8: Predicted induction time vs trapping ratio for different peroxide fractions

Based on previous research, a trapping ratio of 0.75 is optimal for PP crosslinking with AOTEMPO, where 75% percent of initiator-derived radicals (by L-231 + DCP) are quenched by nitroxyl, leaving 25% to initiate crosslinking by oligomerization of pendant acrylate functionality [27, 28]. To this end, the induction time at 0.75 TR for different values of PF has been calculated for comparison to experimental data.

2.5.2. PP functionalisation

A simple analysis of the stoichiometry of an AOTEMPO formulation yields an estimate of the average number of alkoxyamine groups per PP chain. Since only those PP chains that contain polymer-bound acrylate can crosslink efficiently, this number can shape expectations on the amount of gel that can be expected of a PP + AOTEMPO formulation. As such, an

examination of a PP modification reaction involving a nominal DCP loading of 2 wt% and a trapping ratio of 0.75 is worthwhile.

Given that the number-average molecular weight (M_n) of the PP used in this study was 97,000 g/mol, the average concentration of PP chains in a sample corresponded to

$\frac{1}{97000 \text{ g/mol}} \times 10^6 = 10.3 \frac{\mu\text{mol of chains}}{\text{g-PP}}$. That is, in each gram of PP there was an average of 10.3 μmol of polymer chains.

A DCP loading of 2 wt% corresponds to a peroxide concentration of $[\text{ROOR}] = 74 \mu\text{mol/g-PP}$. Therefore, an AOTEMPO formulation with a trapping ratio of 0.75 would have a nitroxyl concentration of $74 \mu\text{mol/g-PP} / 2 * 0.75 = 111 \mu\text{mol/g-PP}$. Note that AOTEMPO traps both PP macroradicals as well as the methyl radicals produced by cumyloxy radical fragmentation. The fraction of AOTEMPO that becomes polymer-bound can be estimated from the H-atom abstraction efficiency of the peroxide acting on PP at 165°C. This is approximately 0.37 [10], meaning that approximately 37% of alkoxy radicals abstract a hydrogen atom from the polymer, while the remaining 63% fragment to a methyl radical + acetophenone. Therefore, of the 111 $\mu\text{mol/g-PP}$ of AOTEMPO charged to a PP formulation, only 37%, or 41 $\mu\text{mol/g-PP}$ is expected to become polymer bound.

Using these values, the average number of AOTEMPO groups bound to each polymer chain =

$\frac{41 \mu\text{mol/g-PP}}{10.3 \mu\text{mol/g-PP}} = 4.0$ groups per PP chain. This degree of functionalization can be expected

to produce a PP thermoset [13]. However, it should be noted that smaller chains in the molecular weight distribution are less likely to be functionalized, and chain scission occurs

concurrently with acrylate oligomerization during the crosslinking phase of an AOTEMPO + PP cure. Both factors will reduce the crosslink density / gel content generated by a cure formulation.

2.5.3. Storage Modulus

The dynamics and extent of polymer cross-linking are typically monitored using time-resolved measurements of the dynamic storage modulus (G') at constant temperature, frequency, and strain amplitude [16]. The data plotted in Figure 9 illustrate the evolution of G' for a series of AOTEMPO + PP formulations of differing peroxide fraction (PF). Replicate runs were conducted for each PF to ensure reproducibility. For clarity, only representative curves for each PF have been shown. Although the AOTEMPO trapping ratio was fixed at 0.75 and the peroxide concentration was fixed at $74 \mu\text{mol/g}\cdot\text{PP}$, the induction time and final storage modulus observed for each formulation was strongly influenced by the relative amounts of DCP and L-231 in the mixture.

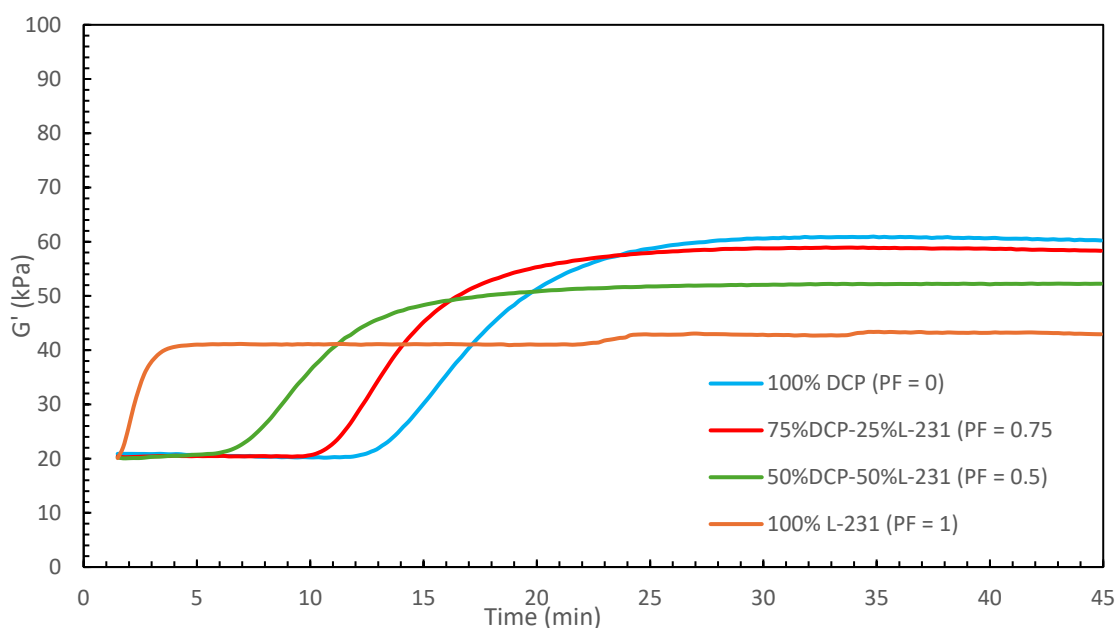


Figure 9: Storage modulus dynamics for various formulation ratios of peroxides used. (DCP = 2.0 wt%, L-231 = 2.2 wt% [$74 \mu\text{mol/g}$], AOTEMPO = 0.75 TR [$111 \mu\text{mol/g}$])

The initial period of constant G' is the observed induction period, t_{ind} , which ends at the point when all nitroxyl functionality is converted to alkoxyamine. [21] At this point the remaining radicals initiate the oligomerisation of pendant acrylate groups for crosslinking, and the cleavage of tertiary radicals for chain scission, creating a new chain architecture based on the balance of these two reactions. A final phase of unfettered PP degradation may also occur leading to some extent of cure reversion [28].

The induction times observed from the G' plot are slightly greater than theoretical values, which can be attributed to challenges in achieving precise reagent loadings. Nevertheless, the underlying concept of the reduction in t_{ind} due to the introduction of L-231 in different quantities is confirmed. That is, there is a clear reduction in induction time as the proportion of L-231 in the dual-initiator mix increases.

Unfortunately, the ultimate degrees of crosslinking observed at the same molar concentration of L-231 and DCP differed substantially (Figure 9), with the faster-decomposing initiator providing inferior performance. This is attributed to the method of sample preparation, which involved solution-coating PP powder with peroxide and AOTEMPO and evaporating off the solvent. This procedure places the reagents on the surface of PP particles, with limited potential of their reaching the interior. Upon charging to a rheometer cavity that is preheated to 165 °C, initiator decomposition and particle melting/reagent diffusion occur concurrently. The half-life of L-231 at this temperature is 0.3 minutes, meaning that a significant amount of this peroxide is expected to break down within the melting period, thereby reducing its ability to affect polymer modification. As a result, PP crosslinking formulations containing a high proportion of L-231 relative to DCP produce lower crosslinking extents. However, a low peroxide fraction of PF=0.25 resulted in

only a slight reduction in G' value, providing a means of controlling induction time without compromising crosslink yields to a great extent.

2.5.4. Gel Content Measurements

The gel content of a polymer sample is defined as the weight percent of insoluble material retrieved from boiling xylene [4].

$$\text{Gel Content (\%)} = \frac{\text{mass of insoluble polymer}}{\text{initial sample mass}} \times 100\% \quad (14)$$

As a measure the fraction of polymer that is crosslinked to the point of effectively infinite molecular weight, it only distinguishes between soluble and insoluble components of a sample. While it provides little insight into the distribution of crosslink density within a cured PP sample, it is a widely reported property of crosslinked materials and, as such, has been measured in the present work. Note that the error in gel content determinations is $\pm 5\%$ based on replicate measurements.

Gel content data for all DCP + L-231 formulations (Table 1) prove that AOTEMPO is capable of generating a thermoset network within a PP homopolymer, with gel fractions in the range of 60% in all samples. This suggests that gel content provides only a general representation of crosslink yield without capturing differences in network architecture. The corresponding storage modulus measurement in the previous section revealed that G' or network stiffness decreases with increasing L-231 concentration. Hence, gel content paired with G' measurement provides a more comprehensive insight into crosslink quality. The most attractive formulation had a PF=0.25 (75% DCP and 25% L-231), which provided a useful induction time and high crosslink density, as indicated by both G' and gel content measurements.

Table 1: Induction time and Gel content readings for different PF formulations of PP ([AOTEMPO] = 111 $\mu\text{mol/g}$)

Sample	Peroxide Fraction (PF)	Peroxide Concentration ($\mu\text{mol/g}$)		Calculated Induction Time (mins)	Experimental Induction Time (mins)	Crosslinked PP Gel content ($\pm 5\%$)
		DCP	L-231			
1	0.00	74	0	6.8	9.8	60%
2	0.25	55.5	18.5	5.4	8.2	62%
3	0.50	37	37	3.4	3.9	62%
4	1.00	0	74	0.6	0.3	60%

2.6. Conclusion

PP thermosets generated with AOTEMPO formulations can provide control of both induction time and crosslink density when mixed peroxide initiators (DCP and L-231) are employed. However, due to the rapid decomposition of L-231 at the temperatures required to process PP, careful consideration must be made regarding the state of mixedness in the starting formulation. The solution-coating procedure used in this work provides a non-uniform mixture composition with peroxide located in the surface region of the PP powder. Inefficiencies arise when high local concentrations of L-231 decompose during PP melting, and overall performance may be improved using an alternate formulation processing technique.

Chapter 3

Ultraviolet (UV) radiation-initiated modification of polymers

Light-induced or photochemical processes offer exceptional control over polymer modification and architecture, where the initiation rates governed by light intensity provide both spatial and temporal precision [35]. Light-driven reactions facilitate the formation, degradation, and crosslinking of polymers, which is utilised across a broad spectrum of applications, from microfabrication to *in vivo* biomedical systems.

Light-based processes offer several distinct advantages in polymer chemistry. Firstly, photochemical initiation provides access to highly energetic states that enable reactions such as bond cleavage, isomerization, and crosslinking to proceed rapidly at ambient conditions, without the need for elevated temperatures or harsh reagents. Another advantage is excellent temporal control, as reactions can be precisely started or halted by light exposure, allowing solvent-free and energy-efficient operation under mild conditions. Finally, photochemical techniques enable spatial control of reactions through patterned or focused illumination, facilitating two- and three-dimensional reaction control of polymer materials for applications ranging from lithography to additive manufacturing [36].

The basic principle of photo-initiated (UV) crosslinking is the utilisation of a reactive species (radical or ions) to initiate the transformation of a multi-functional monomer to a crosslinked polymer (Fig 10). Under UV irradiation the photo-initiator in a formulation generates reactive species which promote network formation through polymerization or crosslinking [37]. Generally, a light-curable formulation contains the following components: an oligomer which is the multifunctional monomer whose backbone determines the property of cured

material; photo-initiators that absorb UV light and are chosen according to the wavelength of radiation and preferred chemistry; a reactive diluent such as a low viscosity monomer that reduces the viscosity of the formulation, and additives such as fillers, inhibitors, pigments, stabilisers, etc [38]. Note that, in case of solid film formulations, reactive diluents are not required.

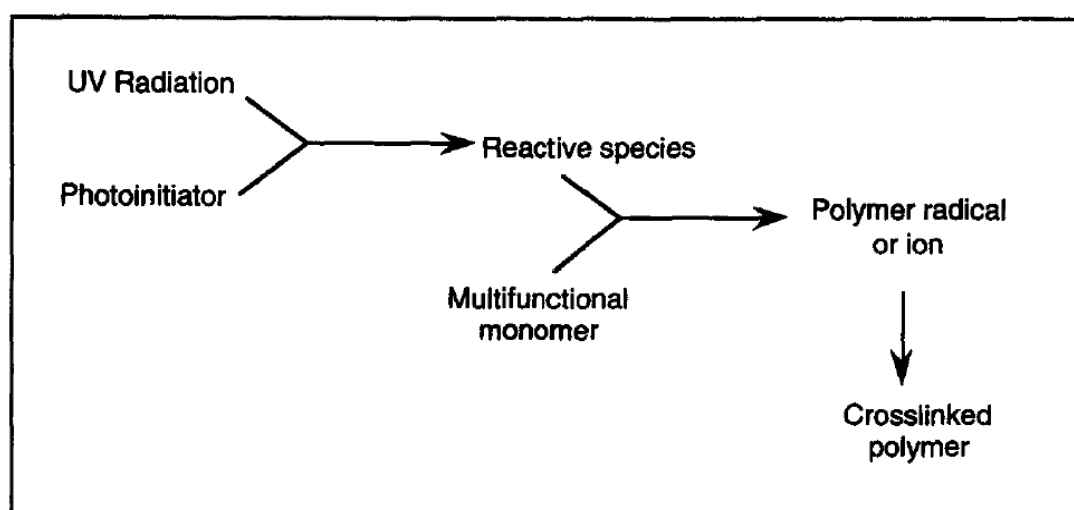


Figure 10: Basic principle of UV crosslinking of polymers

[37]

Since the crosslinking process involves photochemistry that is induced by light absorption, it is governed by two fundamental principles: the Grotthus-Draper Law, which states that chemical reactions occur only when molecules absorb light of sufficient energy, and Einstein's photochemical equivalent law, which establishes that each absorbed photon activates a single molecule. As molecules absorb a photon of sufficient energy, they transition to electronically excited states [39]. The excited molecules might dissipate their energy through various pathways, which is illustrated by a Jablonski Diagram (Fig 11) [40]. These pathways include photodissociation from excited singlet or triplet state by intersystem

crossing; radiative decay producing fluorescence emission (from singlet states) or phosphorescence (from triplet states); or the excitation energy may be transferred to neighbouring molecules [39]. These photophysical processes are in competition with photochemical processes such as isomerisation, bond-breaking and bond-forming reactions. On absorption of UV light, photo-initiators are excited to a singlet state and can undergo intersystem crossing to a triplet state. The highly reactive triplet follows one of the two pathways according to the radical photo-initiating system: a Norrish type I process, which involves unimolecular bond cleavage to generate two radicals, or a Norrish type II process, where radicals are generated when the excited molecule interacts with a second molecule (an example is H-atom abstraction from a co-initiator) [36, 41].

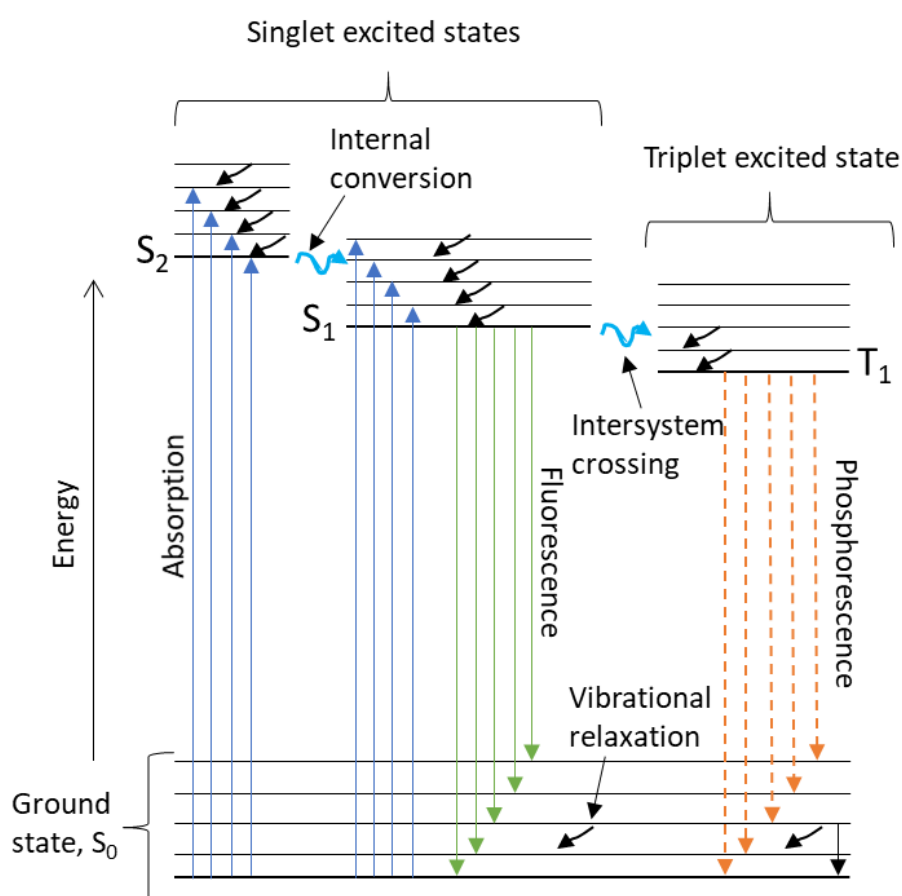


Figure 11: Jablonski Energy Diagram

[40]

Crosslinking by photo-initiation follows a similar mechanism as that by thermal initiation (*via* peroxides) and involves initiation, propagation, chain transfer and termination[42]. This is depicted in Fig.12. Initiation involves the generation of radicals from the photo-initiator by electron transfer or through the pathways described above. These radicals attack monomers or oligomers that contain reactive sites such carbon-carbon double bond (C=C). Consecutive covalent bonds are formed between the oligomers as new radicals are generated, propagating the reaction and forming crosslinks. Some propagating chains may abstract weakly bonded atoms like hydrogen which can lead to their termination as a dead polymer chain and the generation of a new radical which can start propagating. This process is called chain transfer. Finally, termination can occur by recombination of two propagating chains or by disproportionation [43, 44].

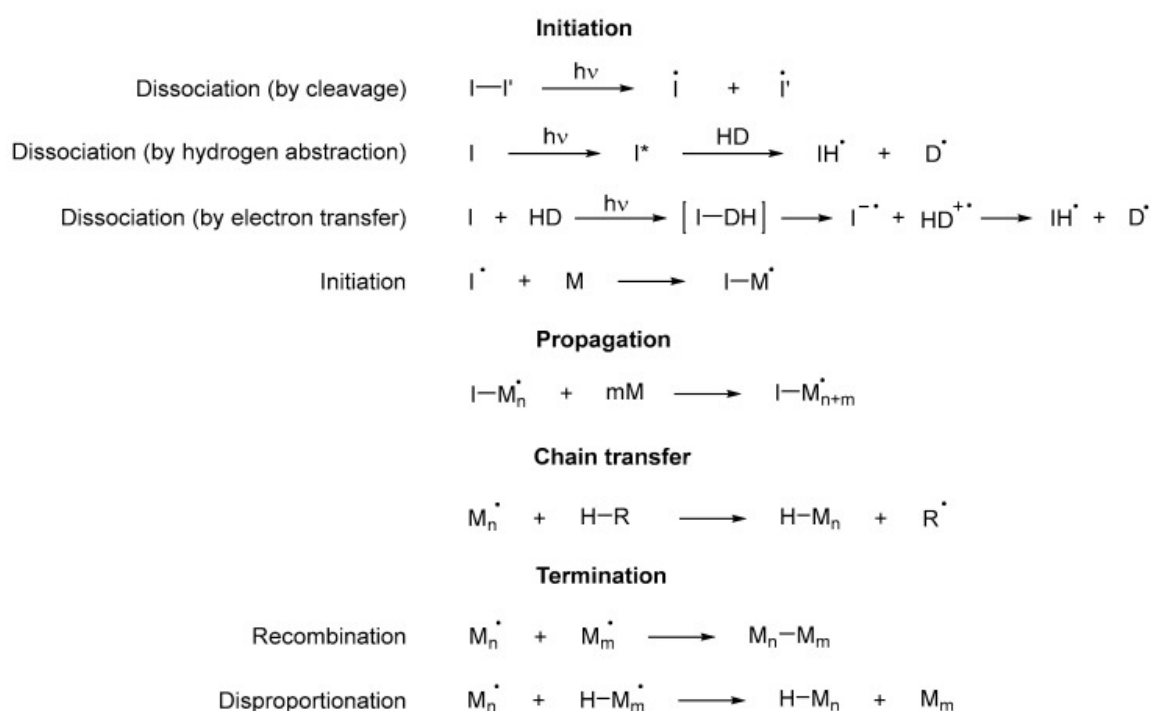


Figure 12: Mechanism of free-radical crosslinking photopolymerization (I- Photo-initiator, D- donor, M- Monomer) [43]

In addition to these free-radical photopolymerization processes there is often another step called inhibition. This is commonly due to molecular oxygen which reacts with growing polymer chains to form peroxy radicals ($\text{POO}\cdot$) that are less reactive and effectively act as dead chains. Practically, oxygen inhibition can introduce an induction period or slow the polymerization near the surface and may lower overall conversion. [45, 46]. In some cases, inhibitory compounds are intentionally introduced to regulate radical propagation.

Nitroxides like TEMPO are commonly used as inhibitors in photopolymerization reactions [47, 48].

3.1. Photo-initiators and Acrylates

Typically, photo-initiators are classified as either Type I or Type II. The mechanism of Type I photo-initiators involves photolytic α -cleavage to yield two radicals. Most Type I photo-initiators are substituted aromatic carbonyl compounds such as analogues of benzoin, acetophenone, and acylphosphine oxides [42]. They have high quantum efficiency and are used extensively in industry, but a drawback is their irreversible consumption during photopolymerization [49]. 2,4,6-trimethylbenzoyldiphenyl phosphine oxide or Lucirin TPO (Fig 13) is a type I photo-initiator based on acylphosphine oxide and is commercially used in curing composite dental resins and printing inks [50, 51].

The activation of Type II photo-initiators (or hydrogen-atom abstraction type initiators) follows a bimolecular mechanism of H-atom abstraction commonly from a co-initiator like a tertiary amine. Their curing rates are slower compared to Type I that follow unimolecular radical generation [52]. Benzophenone, camphor quinone, and thioxanthone are common Type II initiators. When these compounds undergo photolytic cleavage in the presence of a hydrogen-atom donor, they produce a ketyl radical along with another radical derived from

the hydrogen donor. The ketyl radicals are relatively unreactive due to stabilization effects and typically do not participate in the polymerization process [53]. Due to their efficiency, the focus of this research will be on the use of Type I initiators.

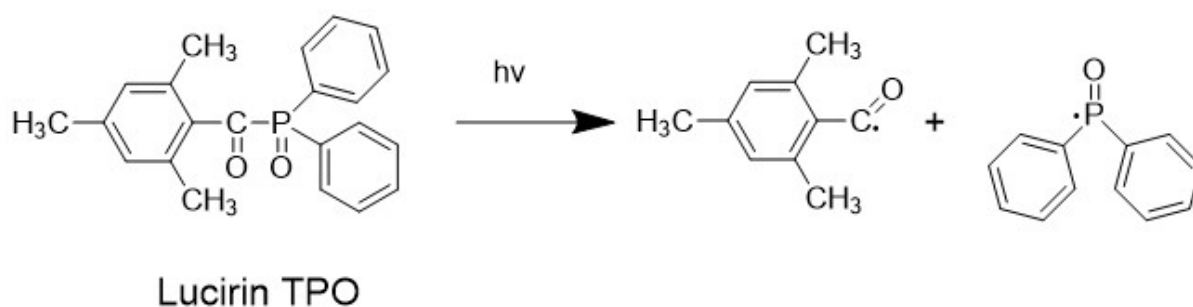


Figure 13: Chemical structure of Lucirin TPO and its photolysis pathway

Acrylate monomers are widely employed in radical photopolymerization with the general mechanism illustrated in Fig. 14. Multifunctional acrylates and methacrylates can yield highly crosslinked networks with superior mechanical, thermal, optical and chemical properties. They are widely used in adhesives, coatings, and complex macromolecular designs [54]. Typically, in UV-curing by a radical mechanism, C=C bonds in acrylate oligomers react with radicals generated by the photo-initiator to create a crosslinked polymer network [55].

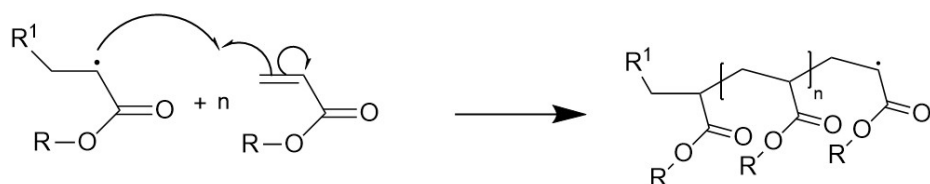
The grafting of acrylate functionality onto polymer backbones or substrates is a well-established method of introducing reactive sites (C=C) that can be photopolymerized to generate a crosslink network. Considerable research has demonstrated polymer functionalisation with acryloyl groups to develop photocurable materials for applications like 3D printing and coatings [56, 57]. The introduction of an acrylate moiety enables the photo-crosslinking of polymers that are conventionally conducted with thermal initiation. Du *et al.*

[58] synthesized a polysiloxane with multiple acryloyl groups that was cured using a mixture of the photo-initiators Irgacure 184 and Irgacure 1173. Grafting of hydrogels with acryloyl or methacryloyl functionality and photopolymerizing them enables precise control of crosslinking and subsequent material properties. An example is the research by Kamoun *et al.* [59] which explored the grafting of poly(vinyl alcohol) (PVA) with glycidyl methacrylate (GMA) to synthesize PVA-g-GMA hydrogel that was photo-crosslinked by the Type I photo-initiator Irgacure 2959.

Initiation



Propagation



Termination

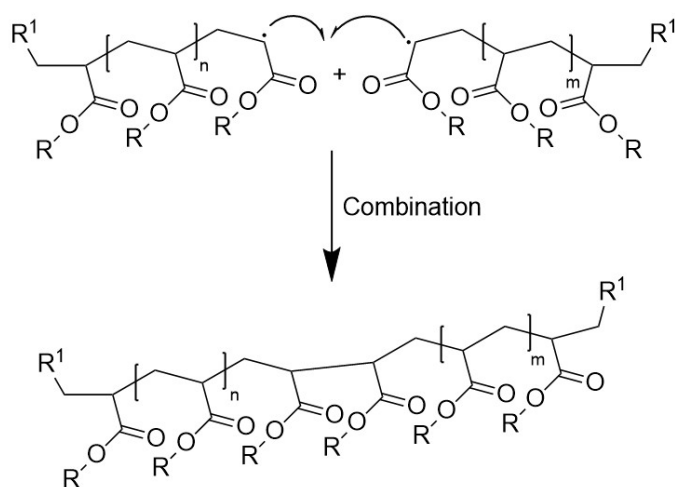


Figure 14: Simplified free radical polymerization mechanism for acrylates ($R^{1\cdot}$ is the radical species formed from the initiator)

3.2. Potential for UV crosslinking of PP

Polypropylene homopolymer thermosets can be prepared by two methods: through thermo-chemical initiators such as the peroxides discussed in Chapter 2, and through high energy radiation. The latter approach involves the use of electron beam (EB) radiation or macroradical generation using photo-initiators that can be activated by UV radiation [15]. In UV radiation-induced crosslinking of PP, some form of chemical species or compound is required to either facilitate hydrogen atom abstraction and/or generate polymer radicals for crosslink formation. An early concept theorized the potential of ash (mainly TiO_2 and Al_2O_3) embedded in PP as a residual catalyst of Ziegler-Natta coordinative polymerization. The components of the ash absorbed UV radiation (in vacuum) at different wavelengths, resulting in the excited ash which collided with polypropylene molecules possibly driving out a hydrogen atom from the tertiary carbon. However, the efficiency was very low as indicated by a gel fraction of just 20% at after 60 hours exposure [60].

Nojiri and Sawasaki [3] explored the use of polyfunctional monomers with acrylate groups ($\text{CH}_2=\text{CH}-\text{COO}-$) to assist in the crosslinking of polypropylene under electron beam and gamma radiation. The acrylate group exhibited remarkably high crosslinking efficiency, achieving gel formation at extremely low doses (0.01-0.05 Mrad or Megarad of absorbed radiation) compared to conventional crosslinking approaches. This is attributed to the high reactivity of acrylate-derived radicals, which efficiently graft onto polymer chains and subsequently form crosslinks through their multiple functional groups. Though this work focused on ionizing radiation, the grafting of acrylate functionality on polypropylene could be utilised for crosslinking via UV radiation.

3.3. AOTEMPO-mediated UV crosslinking of PP

Previous work has explored peroxide-initiated modifications of polyolefins with functional derivatives of TEMPO, with the intent of introducing long chain branching or crosslinking [26, 28]. These nitroxyls prevent premature curing by quenching of polymer macroradicals to yield alkoxyamines in the early stages of the process. Additives such as AOTEMPO not only provide this induction period but promote crosslink density by oligomerisation of pendant acrylate groups [25]. Furthermore, AOTEMPO also traps methyl radicals generated from DCP decomposition, forming methyl alkoxyamines which are incorporated into the polyolefin thermoset. The optimum trapping ratio for PP is on the order of 0.75 as discussed in the previous chapter, which leaves 25% of initiator-derived radicals to oligomerize acrylate functionality to a crosslink network [27].

An alternate approach is explored in this chapter, wherein peroxide initiation of alkoxyamine formation is followed by a successive, UV-initiated acrylate oligomerisation process. Due to their high efficiency, type I photo-initiators have been selected, whose radicals add to the vinyl group in the acrylate functionality to affect network formation. To this end, the maximum increase in the functionalisation of polypropylene chains with pendant acrylate groups is required. Hence, a trapping ratio of 0.99 was chosen, where 99% of peroxide-derived alkyl radicals is quenched to introduce the acrylate moiety, without otherwise affecting the polymer structure. Side reactions such as chain scission, which is a common problem in thermal initiation, are effectively prevented by nitroxyl trapping of alkyl macroradical [28]. The subsequent step involves the irradiation of sample along with a suitable photo-initiator to establish a crosslink network (Fig.15).

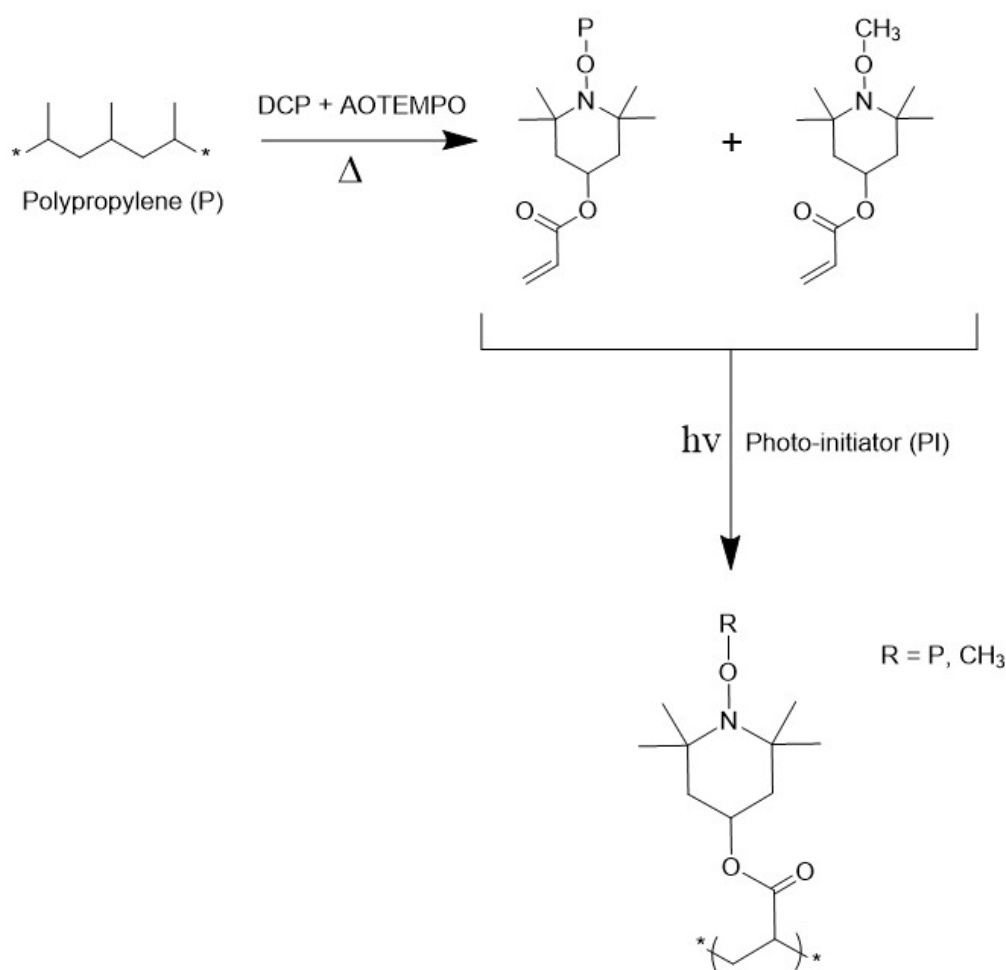


Figure 15: Simplified reaction mechanism for UV crosslinking of PP

3.4. Experimental

3.4.1. Materials

Isotactic polypropylene pellets, dicumyl peroxide (DCP, 98%), acetone and dichloromethane (DCM) were used as received from Sigma-Aldrich. Two grades of PP were used ($M_w=340,000$ g/mol, $M_n=97,000$ g/mol, MFI= 4g/10min) and ($M_w=250,000$ g/mol, $M_n=67,000$ g/mol, MFI= 10 g/10min). 4-Acryloyloxy-2,2,6,6-tetramethylpiperidine-1-oxyl (AOTEMPO) was provided by Ingenia polymers.

Three photo initiators were used as received from Sigma-Aldrich. These included 2-hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone (Irgacure 2959 or I2959), phenylbis(2,4,6-

trimethylbenzoyl)phosphine oxide (Irgacure 819 or I819), and diphenyl(2,4,6-trimethylbenzoyl)phosphine oxide (Lucirin TPO or L-TPO).

3.4.2. PP film preparation

PP pellets were ground to a powder using a tabletop grinding machine. The powdered PP (5g) was divided evenly in glass jars. Standard solutions of the desired reagents DCP and AOTEMPO in acetone were prepared according to the required masses. The powdered PP was coated with these solutions and mixed using a jar mill roller for 15 mins and subsequently dried by exposing to a stream of air. This formulation was then thermally treated at 175°C in the Advanced Polymer Analyzer 2000 (APA, Alpha Technologies) to decompose the peroxide, yielding a PP-grafted macromonomer (as mentioned in Section 3.4.3).

The thermally treated sample was reground and coated with DCM solution of the photo-initiator, mixed and dried as described above, before melt-pressing the sample into a circular film (diameter = 2cm, thickness = 1.1 mm) using a compression moulding machine at 195°C for 30 seconds.

3.4.3. Instrumentation and Analysis

The initial PP formulation was thermally treated in the APA equipped with biconical plates. The operating temperature was 175°C, with a frequency of 1 Hz and a strain amplitude corresponding to a 3° arc.

The circular PP films obtained from compression moulding (195°C, 30 seconds) were exposed to UV light (wavelength = 365 nm) at a light intensity of 95 mW/cm² using an LC8 Hamamatsu Lightning Cure UV light. Both sides of the film were sequentially exposed to UV light for the same duration to ensure uniform UV exposure throughout the film thickness,

since penetration depth in polymer materials is typically limited. UV curing was performed for durations ranging from 1 to 40 minutes. Gel content results are reported up to 30 minutes, since it plateaued after this time duration.

For the determination of gel content, the polypropylene film was cut into smaller pieces of approximately 0.1 g and wrapped in a pre-weighed 120-mesh sieve cloth. The sample was placed in refluxing xylenes stabilized with 100 ppm BHT for 4 hours (ASTM D 2765-01). The sample was retrieved, and residual material was dried in a vacuum oven at 90°C to a constant weight. The difference in final polymer mass compared to initial mass as a percentage provided the gel fraction.

3.5. Results

3.5.1. Degree of PP chain functionalisation at a given M_n and [DCP]

The number average molecular weight (M_n) is critical to the extent of functionalisation

obtained in polypropylene samples. A sample PP of $M_n = 67,000$ has $\frac{1}{67000 \text{ g/mol}} \times 10^6 =$

$14.925 \frac{\mu\text{mol of chains}}{\text{g PP}}$. DCP generates 2 radicals per molecule upon thermal decomposition.

Therefore, a formulation containing 2 wt% DCP corresponds to a concentration of [DCP] = 74 $\mu\text{mol/g}$ that, in turn, generates a total cumyloxy radical concentration of 148 $\mu\text{mol/g}$. Given a H-atom abstraction efficiency of 0.37 for DCP acting upon PP [12], the number of PP macroradicals generated from this peroxide loading is $148 \times 0.37 = 54.76 \mu\text{mol/g}$.

Using an AOTEMPO trapping ratio of $TR = 0.99$ gives [AOTEMPO] = 146.3 $\mu\text{mol/g}$, which is expected to generate a PP-bound AOTEMPO content of $146.3 \mu\text{mol/g} \times 0.37 = 54.13 \mu\text{mol/g}$.

The number of AOTEMPO-derived alkoxyamine groups per PP chain is, therefore, $\frac{54.131}{14.925} =$

3.64 groups per PP chain.

A formulation prepared from a PP with a number average molecular weight of $M_n = 97,000$

has $\frac{1}{97000 \text{ g/mol}} \times 10^6 = 10.31 \frac{\mu\text{mol of chains}}{\text{g PP}}$, with a corresponding alkoxyamine content of

$\frac{54.131}{10.309} = 5.25$ groups per PP chain. As such, it is advantageous to use a PP parent material

of high molecular weight when crosslink density is a desired reaction outcome.

The amount of DCP in the PP formulation also affects the extent of PP functionalization and, consequently, the attainable crosslink density. When 1 wt% DCP with an AOTEMPO trapping ratio of $TR = 0.99$ is used, $[DCP] = 37 \mu\text{mol/g}$. Then the total radical concentration = $74 \mu\text{mol/g}$ and with a H-atom abstraction efficiency of 0.37, the anticipated alkoxyamine concentration is $27.38 \mu\text{mol/g}$. For a PP parent material with $M_n = 67,000$ the number of alkoxyamine groups per chain = $\frac{27.4}{14.925} = 1.82$ groups per PP chain, while a material with $M_n = 97,000$ can provide $\frac{27.4}{10.309} = 2.62$ groups per PP chain. The number of alkoxyamine groups available is much lower in comparison to 2 wt% DCP and hence the extent of crosslinking is expected to be lower. Increasing DCP loading beyond 2 wt% results in poor mixing and uneven coating on PP substrate, yielding no significant improvement.

3.5.2. PP Macromonomer Synthesis

The synthesis of PP macromonomers requires the activation of polypropylene chains through radical chemistry. Thermal treatment is required to activate a peroxide such as DCP in the presence of AOTEMPO to generate a PP-derived macromonomer capable of photo-oligomerization. The results discussed in this section are similar for both grades of PP utilised in the study ($M_n = 67,000 \text{ g/mol}$ and $M_n = 97,000 \text{ g/mol}$). Replicate runs were conducted for all cases shown below to ensure reproducibility. For clarity, only representative curves have been shown.

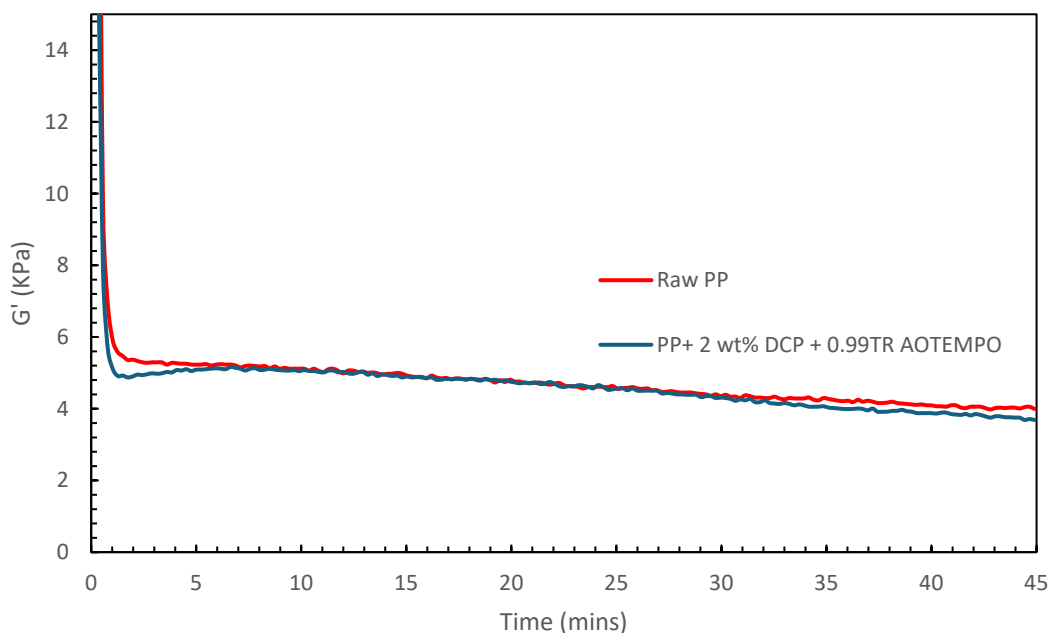


Figure 16: Dynamic storage modulus vs time for PP at 175°C ([DCP] = 74 $\mu\text{mol/g}$, [AOTEMPO] = 146.4 $\mu\text{mol/g}$)

The thermal treatment of PP formulations containing DCP with AOTEMPO at a trapping ratio of 0.99 in the APA enabled rheological analysis via monitoring of storage modulus (G').

Figure 16 illustrates the G' of the PP parent material (raw PP) and a PP+DCP+AOTEMPO formulation as a function of time. As indicated by the stable G' versus time profile, the polymer architecture did not change appreciably when a TR = 0.99 was applied, as nearly all DCP-derived radicals were trapped efficiently by the nitroxyl. The timeframe required for functionalisation of PP chains is dictated by the half-life of the peroxide, with 6-10 half-lives being employed for most crosslinking processes. A thermal treatment time of 7.6 minutes at 175°C was used for all PP macromonomer preparations, which corresponds to six half-lives of DCP at this temperature ($t_{1/2} = 1.3 \text{ min}$), ensuring $\geq 98\%$ peroxide decomposition [61]. A short period is optimal, since prolonged exposure to high temperature may lead to thermal

decomposition of alkoxyamines and degradation of polypropylene via β -scission, which would reduce the efficiency of macromonomer formation [28].

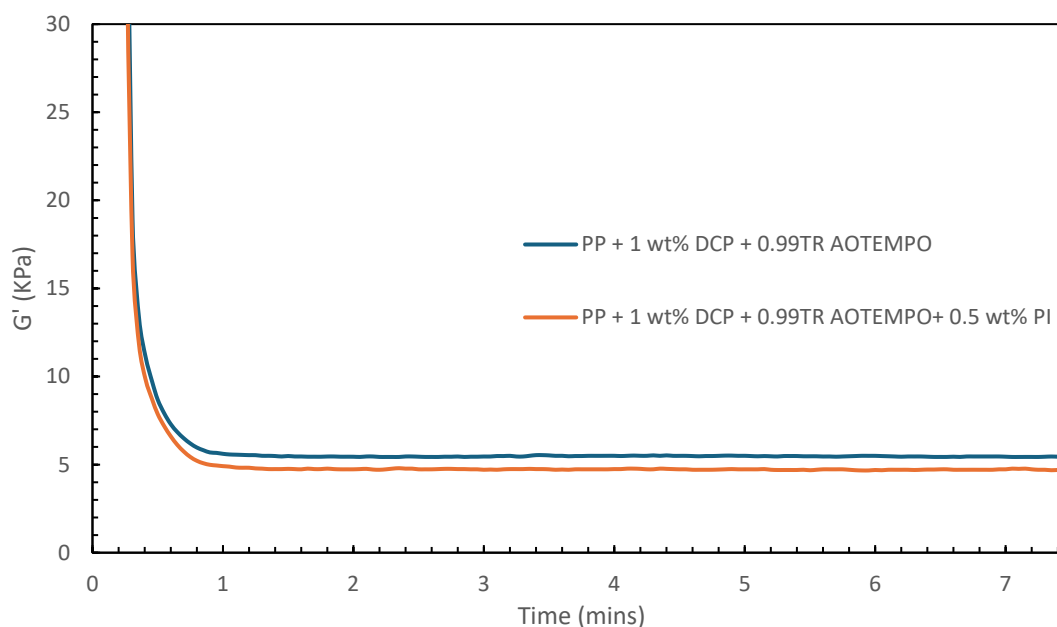


Figure 17: Dynamic storage modulus of PP treated with 1 wt% DCP, 0.99 TR AOTEMPO & 0.5 wt% Photo-initiator (PI) at 175°C ([DCP] = 37 $\mu\text{mol/g}$, [AOTEMPO] = 73.2 $\mu\text{mol/g}$)

Initial trials involved the addition of photo-initiator (PI) after polypropylene was converted to a macromonomer by reaction with DCP + AOTEMPO at high temperature. This PP derivative was ground to a powder before adding photo-initiator by solution-coating, drying, and melt pressing to produce a thin film. Later trials investigated whether similar results could be obtained when the photo-initiator was added along with the PP formulation (containing DCP and AOTEMPO), exposing both initiators to the PP macromonomer preparation conditions. The dynamic storage modulus data observed during macromonomer production was similar for both cases ($G' \approx 5$ kPa), with no significant difference between them.

The storage modulus plots including 0.5 wt% of photo-initiator has been reported for 1 wt% DCP + 0.99TR AOTEMPO in Fig. 17 and 2 wt% DCP + 0.99TR AOTEMPO in Fig. 18. Hence, it was evident that the photo-initiators were not affected by thermal initiation and did not contribute to any significant change in storage modulus during this step. The crosslink yield after UV exposure for both cases was assessed in terms of gel content, as shown in section 3.5.3.

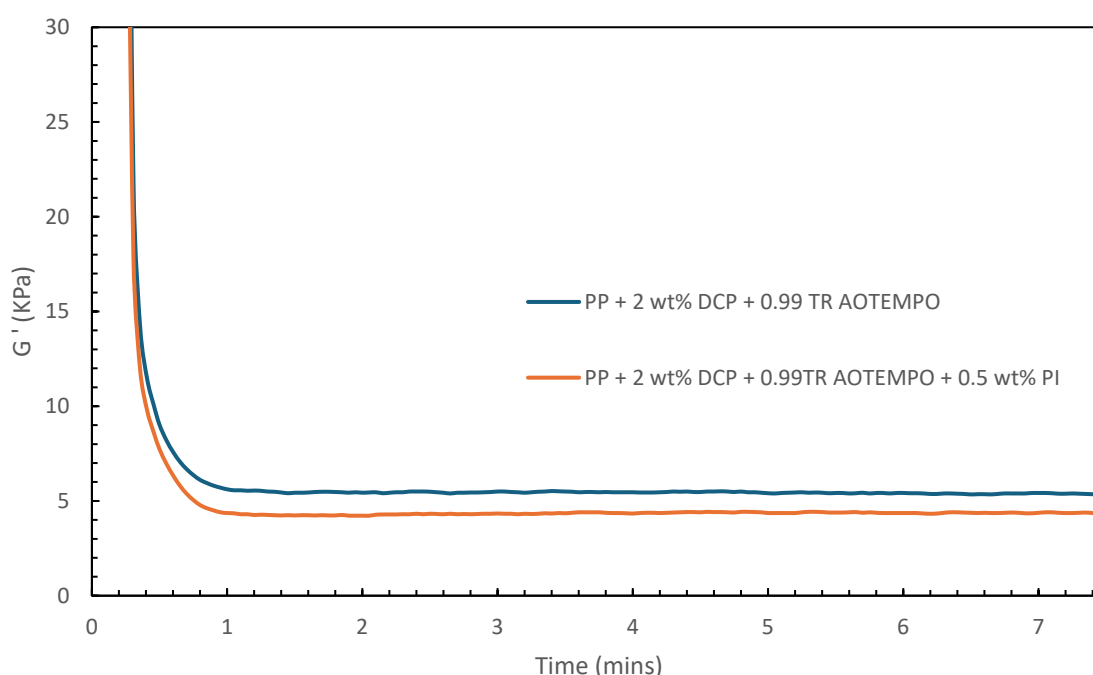


Figure 18: Dynamic storage modulus of PP treated with 2 wt% DCP & 0.99 TR AOTEMPO & 0.5 wt% Photo-initiator (PI) at 175°C ([DCP] = 37 $\mu\text{mol/g}$, [AOTEMPO] = 73.2 $\mu\text{mol/g}$)

3.5.3. Gel Content Measurements

Peroxide-grafted PP-g-AOTEMPO macromonomers were melt pressed into circular films of 1 mm thickness and 2 cm diameter. The UV crosslinking of PP-g-AOTEMPO macromonomers was carried out by irradiating both sides of the circular film with a 365 nm wavelength and a maximum stable intensity of 95 mW/cm² for the same duration. Initial experiments were

conducted using I2959 as the photo-initiator, followed by trials using I819 and L-TPO. Unless otherwise specified, all UV curing runs were carried out with PP-67 K ($M_n = 67,000$ g/mol), hereafter referred to as PP.

Reference experiments were carried out to determine whether type I photo-initiators (I2959, L-TPO, and I819) alone can photo-initiate crosslinking without prior generation of a macromonomer. Table 5 provides gel content measurements of peroxide-free PP films containing 146.4 AOTEMPO $\mu\text{mol}/\text{gram-PP}$ after irradiation for up to 20 min. A lack of a significant gel fraction confirmed the inability of these photo-initiators to activate a PP + AOTEMPO cure formulation. As such, peroxide-initiated grafting of AOTEMPO to PP is required to support a UV-curing process.

Table 2: Gel Content data for UV irradiated PP films containing only AOTEMPO and 0.5 wt% of different Photo-initiators ([AOTEMPO] = 146.4 $\mu\text{mol}/\text{g}$, [I2959] = 22.3 $\mu\text{mol}/\text{g}$, [I819] = 12 $\mu\text{mol}/\text{g}$, [LTPO] = 14.4 $\mu\text{mol}/\text{g}$).

Time	Gel Content - I2959	Gel Content - L-TPO	Gel Content - I819
1 min	2%	2%	2%
3 mins	2%	2%	2%
5 mins	2%	4%	3%
10 mins	2%	4%	3%
20 mins	2%	4%	3%

All gel content measurements reported in this work were performed in duplicate for each UV exposure time. Reported values represent the mean of two independent trials, with variability expressed as the pooled Standard Error (SE) calculated across all UV exposure

times for each sample. The SE across all measurements (n=2) ranged from 0.64 to 1.57%, indicating good reproducibility.

3.5.3.1. Comparison of Photo-initiator addition strategies

Section 3.5.2 described two different approaches for incorporating the photo-initiator (PI) into the UV-curable polypropylene formulation. To compare the crosslinking efficiency of these two methods, gel content testing was performed.

Case 1 (post-thermal treatment addition of PI): The polypropylene formulation containing 1 wt% DCP and 0.99TR AOTEMPO was compression moulded in the rheometer at 175°C to produce the macromonomer derivative. After this thermal processing step, the product was reground. The photo-initiator I2959 was then added, followed by a final melt-pressing stage at 195°C for approximately 30 seconds to create the photocurable polypropylene film. The film was then exposed to UV irradiation. The same process was repeated for a DCP concentration of 2 wt% and corresponding 0.99 TR AOTEMPO content.

Case 2 (pre-thermal treatment addition of PI): In this approach, the photo-initiator I2959 was incorporated into the polypropylene formulation along with DCP and AOTEMPO before thermal treatment. The complete formulation was then processed together at 175°C, followed by melt-pressing and subsequent UV irradiation.

The gel content results for both cases are presented in Figure 19. While both approaches showed comparable increases in gel content with UV exposure time, Case 2 yielded slightly higher gel content values from UV exposure periods of 5 to 20 minutes. This trend could be attributed to the longer melt-state residence time experienced by the photo-initiator in Case

2 (7.6 minutes at 175°C followed by 30 seconds at 195°C) compared to Case 1 (30 seconds at 195°C). The extended melt exposure likely facilitated more thorough mixing and dispersion of the photo-initiator within the polymer matrix [62], potentially leading to more uniform photochemical conversion throughout the film thickness upon UV exposure. Additionally, this approach simplifies processing by eliminating the need to reintroduce reagents after thermal treatment. The maximum gel content obtained was approximately 50% after 20-30 minutes of UV exposure.

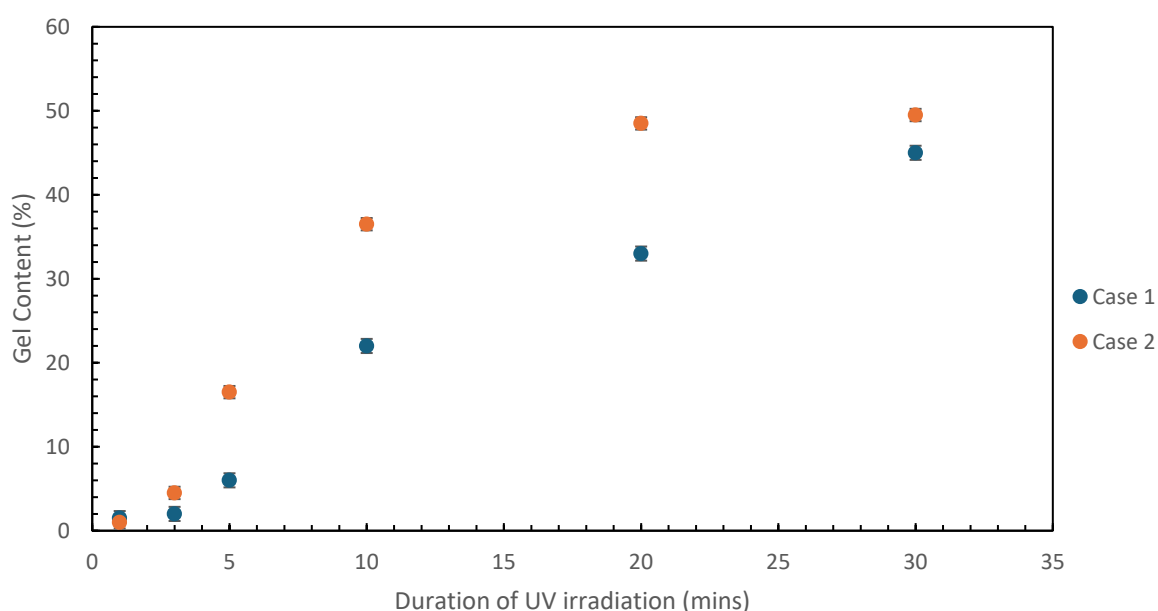


Figure 19: Gel content readings vs. irradiation time for UV cured PP containing 1wt% DCP & 0.99 TR AOTEMPO with 0.5 wt% I2959 charged before (Case 1) and after (Case 2) thermal treatment at 175°C ([DCP] = 37 $\mu\text{mol/g}$, [AOTEMPO] = 73.2 $\mu\text{mol/g}$, [I2959] = 22.3 $\mu\text{mol/g}$). Error bars represent $\pm$ pooled SE (n=2).

All trials included a control experiment involving a gel content test on a melt-pressed film that was not exposed to UV radiation. These control runs yield no significant gel (1%), confirming a lack of crosslinking during macromonomer production. That is, thermal treatment of PP formulations containing DCP, a 0.99 trapping ratio of AOTEMPO, and/or photo-initiator results only in alkoxyamine formation, while significant crosslinking can only

be initiated only during a subsequent UV exposure. Hereafter, all UV curable polypropylene films were prepared with the photo-initiator incorporated during peroxide-initiated macromonomer production and subsequently melt pressed into films.

3.5.3.2. Effect of Photo-initiator Concentration

The effect of photo-initiator concentration on crosslinking efficiency was investigated by comparing formulations containing 0.5 wt% and 1.0 wt% of I2959. This comparison was conducted for two different DCP loadings (1 wt% and 2 wt%, each with the corresponding stoichiometric amount of 0.99TR AOTEMPO). The gel content values as a function of UV exposure time for both photo-initiator concentrations are presented in Fig. 20 (1 wt% DCP) and Fig. 21 (2 wt% DCP).

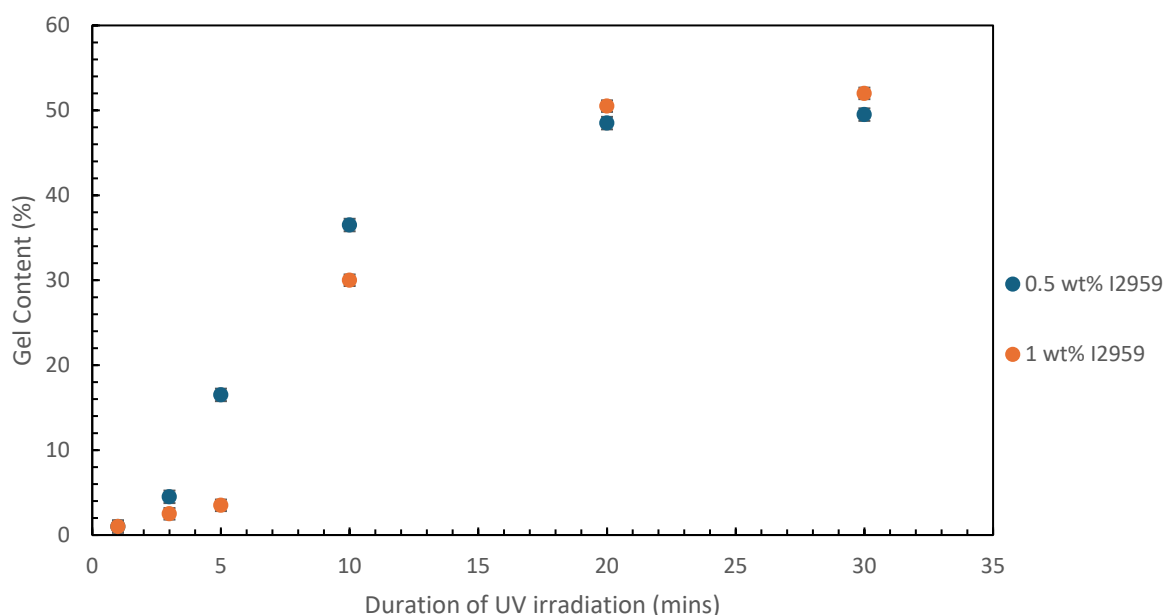


Figure 20: Gel content vs. irradiation time for UV cured PP containing 1wt% DCP & 0.99TR AOTEMPO with two different I2959 concentrations (0.5 wt% [22.3 $\mu\text{mol/g}$] and 1 wt% [44.6 $\mu\text{mol/g}$]); [DCP] = 37 $\mu\text{mol/g}$ and [AOTEMPO] = 73.2 $\mu\text{mol/g}$. Error bars represent $\pm$ pooled SE (n=2).

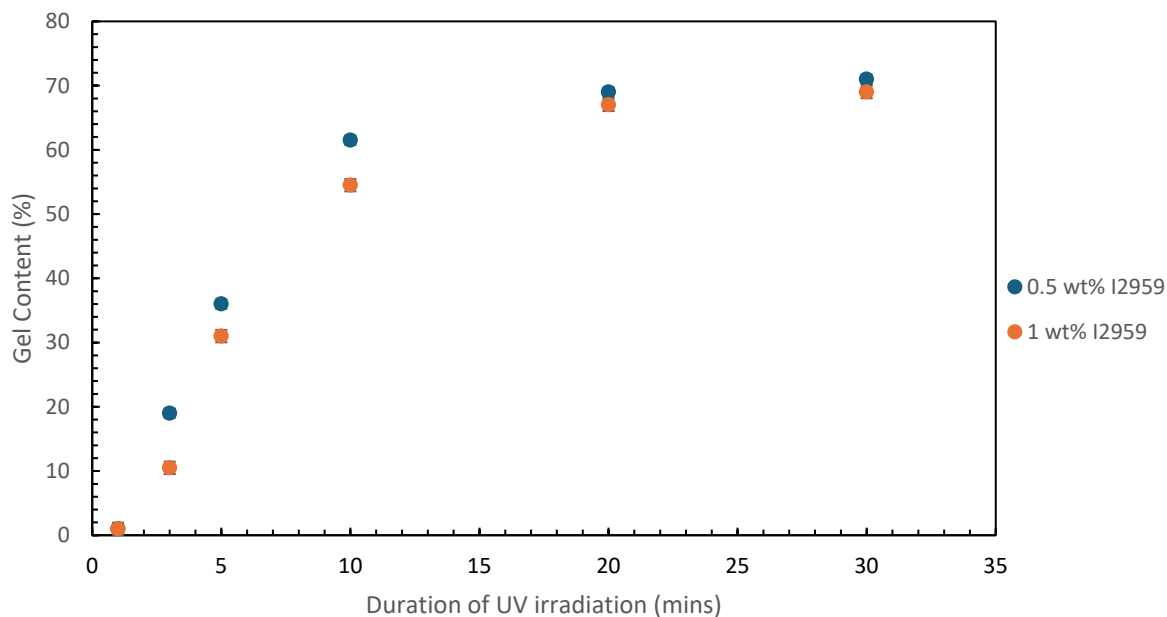


Figure 21: Gel content vs. irradiation time for UV cured PP containing 2 wt% DCP & 0.99 TR AOTEMPO with two different I2959 concentrations (0.5 wt% [22.3 $\mu\text{mol/g}$] and 1 wt% [44.6 $\mu\text{mol/g}$] ; [DCP] = 74 $\mu\text{mol/g}$, [AOTEMPO] = 146.4 $\mu\text{mol/g}$). Error bars represent $\pm$ pooled SE (n=2).

The results showed that increasing the photo-initiator concentration from 0.5 wt% to 1.0 wt% did not significantly enhance the gel content values at equivalent UV exposure times. Both concentrations yielded comparable crosslinking efficiency throughout the exposure period. Based on these findings, 0.5 wt% of I2959 was deemed sufficient for effective photochemical crosslinking of the polypropylene formulation, and this concentration was adopted for subsequent experiments.

3.5.3.3. Effect of DCP & AOTEMPO Loading on crosslinking efficiency

The influence of DCP + 0.99 TR AOTEMPO concentration (for 1 wt% and 2 wt% DCP) on crosslinking yield was evaluated by measuring the gel content of corresponding UV-cured PP-

g-AOTEMPO films as a function of UV exposure time (Fig. 22). The PP films prepared with 2 wt% DCP exhibited higher gel content in comparison to those with 1 wt% across most exposure times. Hence, it is evident from the gel content plots, that increasing DCP concentration (along with the corresponding stoichiometric amount of 0.99TR AOTEMPO) leads to higher extent of alkoxyamine formation. This elevated macromonomer content provides a higher density of polymer-bound pendant acrylate functionality, which promotes more extensive crosslinking upon subsequent UV irradiation. Consequently, the enhanced crosslink density is reflected in higher gel content values.

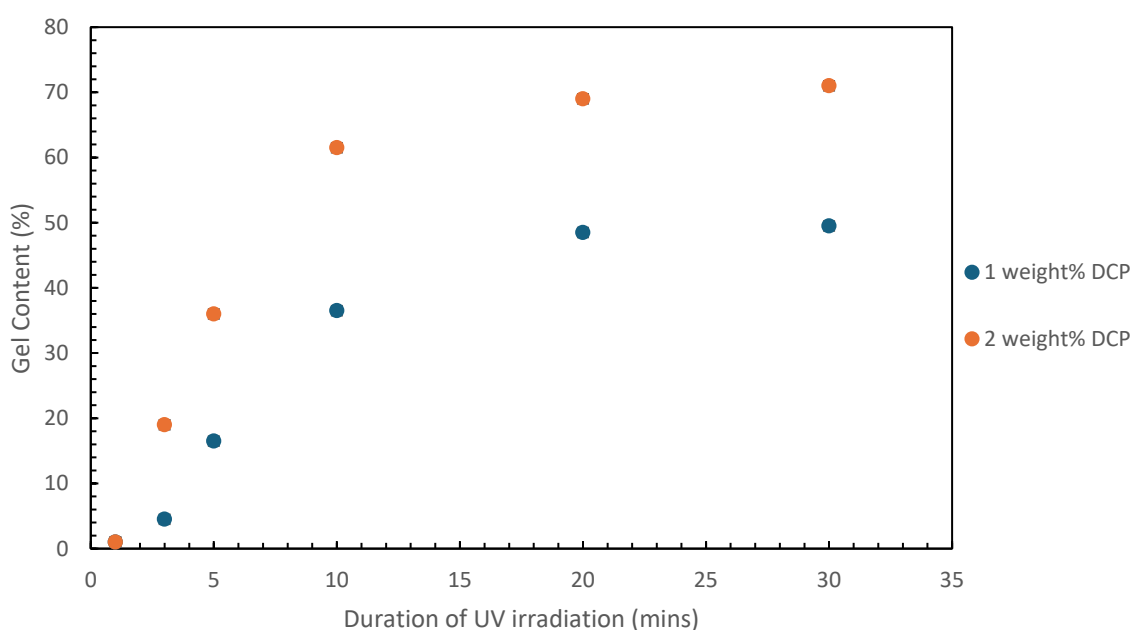


Figure 22: Gel Content vs. irradiation time for PP formulations with two different DCP+AOTEMPO loadings (1 wt% DCP [37 $\mu\text{mol/g}$] + AOTEMPO [73.2 $\mu\text{mol/g}$]; 2 wt% DCP [74 $\mu\text{mol/g}$] + AOTEMPO [146.4 $\mu\text{mol/g}$]) with 0.5 wt% I2959 [22.3 $\mu\text{mol/g}$]. Error bars represent $\pm$ pooled SE ($n=2$).

The gel content increased linearly with UV exposure time for all DCP loadings tested.

Notably, significant gel formation was observed after only 5 minutes of UV irradiation, indicating that this duration is sufficient to establish the initial crosslink network [15].

Further irradiation leads to a substantial increase in the gel content. At 20-30 minutes of UV exposure, the formulation containing 2 wt% DCP achieved a maximum gel content of approximately 70%, compared to ~50% gel content obtained with 1 wt% DCP loading (Fig. 22). This demonstrates the positive correlation between DCP/AOTEMPO concentration and final crosslinking efficiency. Hence, an optimal PP UV curable formulation contains 2 wt% DCP with 0.99TR AOTEMPO and 0.5 wt% PI.

3.5.3.4. Influence of different Photo-initiators and Polypropylene M_n on crosslinking

To evaluate the efficiency of different photo-initiators (PI), the performance of I819 and LTPO was compared to that of I2959. These UV curing runs were conducted using a PP formulation containing 2 wt% DCP + AOTEMPO TR=0.99 with the stoichiometric amount of PI relative to 0.5 wt% I2959. The gel content recorded as a function of UV exposure time for PP-g-AOTEMPO films cured with each photo-initiator is presented in Fig. 23.

All three photo-initiators exhibited similar trends, with gel content increasing progressively with irradiation time. Minor variations in gel content were observed among the three initiators during the initial exposure period (1-10 minutes); however, all formulations reached comparable gel content values by 20 minutes of UV exposure. The final gel content plateaued at approximately 68-70% for all three photo-initiators, indicating that the choice of photo-initiator does not significantly affect the final crosslinking density under the conditions studied.

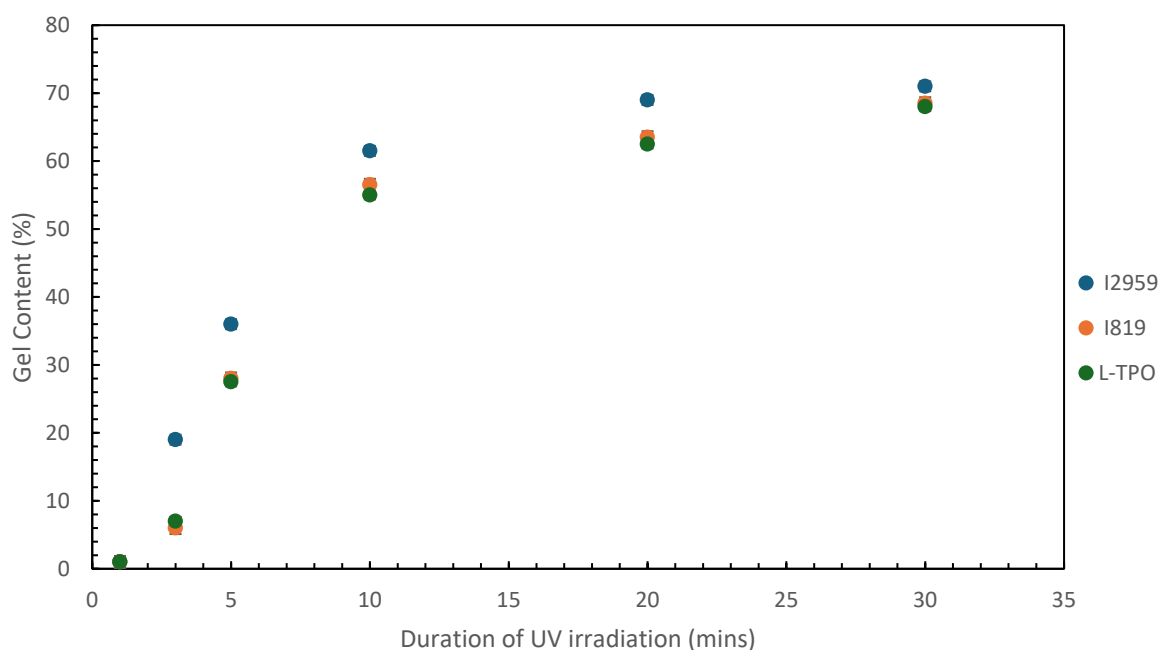


Figure 23: Gel Content vs. irradiation time for UV cure of PP (PP-67k, Mn= 67,000) containing 2wt % DCP & 0.99 TR AOTEMPO with different photo-initiators at 0.5 wt% concentration ([I2959] = 22.3 $\mu\text{mol/g}$, [I819] = 12 $\mu\text{mol/g}$, [L-TPO] = 14.4 $\mu\text{mol/g}$; [DCP] = 74 $\mu\text{mol/g}$, [AOTEMPO] = 146.4 $\mu\text{mol/g}$). Error bars represent $\pm$ pooled SE (n=2 per exposure time).

The influence of polymer molecular weight on photocuring efficiency was investigated using an additional grade of PP with higher molecular weight (PP-97K, Mn = 97,000 g/mol) in comparison to the PP-67K used in all previous photocuring trials. The gel content values for UV cure of PP-97K (with 2 wt% DCP and 0.99TR AOTEMPO) using each of the same initiators (I2959, I819 or LTPO at 0.5 wt%) are shown in Fig 24. In the case of PP-97K, I819 and L-TPO yielded significantly higher gel content during the initial irradiation period (1-3 minutes), while I2959 reached comparable levels by 5 minutes. Overall, the gel content obtained during the early stages of UV irradiation (1-5 minutes) for PP-97K was substantially higher than that observed for PP-67K. This enhanced early-stage crosslinking can be attributed to the longer chain architecture of higher molecular weight polymers, which provides more sites for functionalization. These functionalized longer chains are more readily encountered

by photo-initiated radicals, facilitating faster crosslink formation. Additionally, longer chain lengths result in increased chain entanglements, which act as temporary physical crosslinks and promote more efficient radical propagation, thereby enhancing gel formation during the initial exposure period [63, 64].

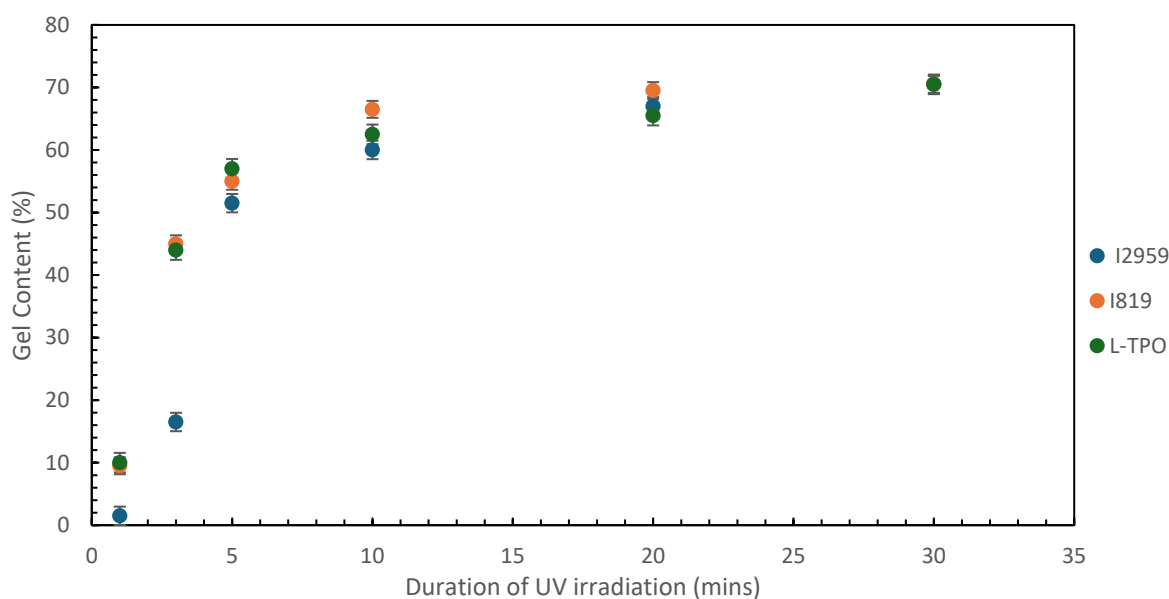


Figure 24: Gel Content vs. irradiation time for UV cure of PP-97k ($M_n = 97,000$) containing 2wt % DCP & 0.99 TR AOTEMPO with different photo-initiators at 0.5 wt% concentration ($[I2959] = 22.3 \mu\text{mol/g}$, $[I819] = 12 \mu\text{mol/g}$, $[L-TPO] = 14.4 \mu\text{mol/g}$; $[DCP] = 74 \mu\text{mol/g}$, $[AOTEMPO] = 146.4 \mu\text{mol/g}$). Error bars represent $\pm$ pooled SE ($n=2$ per exposure time).

The gel content however, plateaued at approximately 70% for both molecular weight grades, consistent with plateau behaviour reported in the crosslinking literature for peroxide-based polyolefin systems [17, 65]. This suggests that maximum gel content is mainly governed by the available acrylate functionality and accessible amorphous fraction. This maximum value likely represents the threshold crosslink density at which nearly all acrylate functionality has reacted, and the crosslinked network has reached completion. At this stage, further chain

mobility and radical diffusion become severely restricted, preventing additional crosslinking reactions.

3.6. Conclusion

The pendant acrylate groups generated through peroxide-initiated grafting of AOTEMPO to PP ($M_n = 67,000$ g/mol) support an efficient secondary crosslinking pathway *via* UV curing. A photo initiator (PI) concentration of 0.5 wt% was sufficient for effective UV crosslinking of PP-g-AOTEMPO macromonomers. Increasing the DCP loading (at a constant TR = 0.99 of AOTEMPO) from 1 wt% to 2 wt% raised the macromonomer content of the functionalized PP derivative, thereby increasing the extent of crosslinking from ~50% to ~70% gel content.

All three photo-initiators evaluated - I2959, I819, and L-TPO - produced significant gel fractions given sufficient irradiation time, demonstrating that acrylate-functionalized photocuring is a viable approach for polypropylene crosslinking. However, under the applied operating conditions, gel content plateaued at approximately 70%. This is attributed to crosslinking being confined to the amorphous phase of PP, as covalent bond formation does not occur within the crystalline region [66]. Although peroxide-initiated grafting proceeds in the fully amorphous melt, recrystallisation excludes the alkoxyamine groups from crystalline domains, restricting UV-induced crosslinking to the amorphous fraction. Quantitative crystallinity data was not obtained in this work since it was not in the scope of the current study. Future work on crystallinity measurements would prove essential to evaluate the impact of crystallinity on maximum crosslink yield.

UV curing of higher molecular weight PP-97K ($M_n = 97,000$ g/mol), exhibited greater initial gel contents and early crosslink development (1-5 minutes) in comparison to PP-67K. However, formulations for both grades of PP converged toward a similar plateau at roughly 70% gel content at longer exposure times (20–30 minutes), representing the maximum crosslink yield value that can be achieved. Although gel content reached a maximum at 30 minutes, there is a possibility that further irradiation can influence network properties like stiffness and network architecture, which may not be captured by gel content alone. Dynamic Mechanical Analysis (DMA) could provide additional insight but could not be effectively performed due to the small dimensions of the film obtained.

Notably, the maximum gel fraction achieved through UV curing (~70%) is higher than that typically obtained through thermal crosslinking alone (~60%). Overall, these findings demonstrate that UV curing provides a robust and effective route for enhancing crosslink density in PP with appropriate processing conditions.

Chapter 4

Conclusions and Future Work

4.1. Conclusions

Prior to the present work, nitroxyl-mediated crosslinking of polyolefins has been extensively explored under thermal processing conditions and perceived as means of scorch protection by introducing a delayed onset of crosslinking. The grafting of AOTEMPO onto the PP backbone enabled a dual curing pathway of thermal initiation via peroxide chemistry and UV-induced cure utilizing the pendant acrylate moiety.

The dual-initiator peroxide chemistry (DCP + L-231) showed the potential to modulate induction time without significant reduction in crosslink yields. This provides a practical means of tuning processing conditions, particularly when long induction times are not required, while maintaining effective cure yields.

The UV cure of acrylate functionalised PP films proved to be successful, yielding high gel content. UV cure proves to be a better option to crosslink PP as it operates at room temperature, preventing chain scission and molecular weight degradation associated with high-temperature peroxide processing. Additionally, this method also offers lower energy consumption and precise special control in contrast to thermal means where the crosslinking occurs in the bulk. These characteristics make UV curing particularly attractive in applications which require enhanced thermal stability, creep resistance and improved mechanical strength, such as food and industrial packaging films, cable insulation and functional coatings.

4.2. Future Work

A comprehensive characterisation of both UV-cured and thermally cured PP samples could help to establish direct comparison of the resulting network structure and properties. DMA performed on UV-cured samples of sufficient dimensions would provide storage modulus that can be compared with the same obtained from rheological measurement for thermally crosslinked PP. This enables an extensive assessment of the relative quality of crosslink density achieved by both methods. Differential Scanning Calorimetry (DSC) is required to compare the crystallinity in both cured and uncured samples, since crystallinity directly influences radical mobility and overall crosslink efficiency.

The UV curing conditions employed in the current work represent only a baseline rather than the maximum achievable performance. The maximum intensity available from the light source used was 95 mW/cm^2 , which is relatively low. A UV source capable of delivering higher intensities would substantially reduce cure times relative to the current results. The trade-off, however, is that beta-scission may occur due to thermal buildup when PP films are subjected to high-intensity irradiation. Extensive investigation would enable determination of the optimum intensity and processing conditions that yield high crosslink densities within a short duration without inducing degradation of the polypropylene matrix. A more thorough understanding of the kinetics governing the photocuring process also warrants further study. Oxygen inhibition is a well-documented barrier to efficient radical-mediated photo-crosslinking at the film surfaces. Strategies to mitigate its effects and other impediments to achieving optimal crosslink yields require dedicated investigation. Potential approaches include curing under an inert atmosphere, or the use of surface barrier coatings to exclude ambient oxygen during irradiation. From an application standpoint, scale-up to

continuous industrial processing represents an important future direction. The transition from static laboratory-scale UV exposure to continuous industrial processing introduces further engineering challenges.

Finally, long-term characterization of the UV-cured films under end-use conditions remains an open area of investigation. Assessment of oxidative stability, resistance to thermal aging, and retention of mechanical performance over time would be essential steps toward evaluating the suitability of this material for applications where durability is a critical requirement.

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Appendix

Appendix A: Raw gel content data for UV-initiated crosslinking of polypropylene

1.1. Comparison of Photo-initiator addition strategies

Table A1: Raw gel content data of UV cured PP containing 1wt% DCP & 0.99 TR AOTEMPO with 0.5 wt% I2959 charged before and after thermal treatment at 175oC ([DCP] = 37 $\mu\text{mol/g}$, [AOTEMPO]= 73.2 $\mu\text{mol/g}$, [I2959] = 22.3 $\mu\text{mol/g}$).

UV Exposure Time (mins)	Case 1 – Post-thermal treatment addition of I2959 (%)			Case 2 – Pre-thermal treatment addition of I2959 (%)		
	Trial 1	Trial 2	Average	Trial 1	Trial 2	Average
1	2.0	1.0	1.5	1.0	1.0	1.0
3	3.0	1.0	2.0	4.0	5.0	4.5
5	5.0	7.0	6.0	17.0	16.0	16.5
10	23.0	21.0	22.0	38.0	35.0	36.5
20	32.0	34.0	33.0	49.0	48.0	48.5
30	45.0	45.0	45.0	49.0	50.0	49.5

1.2. Effect of Photo-initiator concentration

Table A2: Raw gel content data of UV cured PP containing 1wt% DCP & 0.99 TR AOTEMPO with two different I2959 concentrations ([DCP] = 37 $\mu\text{mol/g}$, [AOTEMPO]= 73.2 $\mu\text{mol/g}$).

UV Exposure Time (mins)	0.5 wt% of I2959 (22.3 $\mu\text{mol/g}$) (%)			1 wt% of I2959 (44.6 $\mu\text{mol/g}$) (%)		
	Trial 1	Trial 2	Average	Trial 1	Trial 2	Average
1	1.0	1.0	1.0	1.0	1.0	1.0
3	4.0	5.0	4.5	2.0	3.0	2.5
5	17.0	16.0	16.5	4.0	3.0	3.5
10	38.0	35.0	36.5	31.0	29.0	30.0
20	48.0	49.0	48.5	50.0	51.0	50.5
30	49.0	50.0	49.5	51.0	53.0	52.0

Table A3: Raw gel content data of UV cured PP containing 2wt% DCP & 0.99 TR AOTEMPO with two different I2959 concentrations ([DCP] = 74 $\mu\text{mol/g}$, [AOTEMPO]= 146.4 $\mu\text{mol/g}$).

UV Exposure Time (mins)	0.5 wt% of I2959 (22.3 $\mu\text{mol/g}$) (%)			1 wt% of I2959 (44.6 $\mu\text{mol/g}$) (%)		
	Trial 1	Trial 2	Average	Trial 1	Trial 2	Average
1	1.0	1.0	1.0	1.0	1.0	1.0
3	18.0	20.0	19.0	11.0	10.0	10.5
5	36.0	36.0	36.0	32.0	30.0	31.0
10	61.0	62.0	61.5	53.0	56.0	54.5
20	68.0	70.0	69.0	66.0	68.0	67.0
30	70.0	72.0	71.0	68.0	70.0	69.0

1.3. Effect of DCP+AOTEMPO Loading on crosslinking

Table A4: Raw gel content data of UV cured PP containing two different DCP+AOTEMPO loadings with 0.5 wt% I2959 ([I2959] = 22.3 $\mu\text{mol/g}$).

UV Exposure Time (mins)	1 wt% DCP [37 $\mu\text{mol/g}$] + 0.99TR AOTEMPO [73.2 $\mu\text{mol/g}$] (%)			2 wt% DCP [74 $\mu\text{mol/g}$] + 0.99TR AOTEMPO [146.4 $\mu\text{mol/g}$] (%)		
	Trial 1	Trial 2	Average	Trial 1	Trial 2	Average
1	1.0	1.0	1.0	1.0	1.0	1.0
3	3.0	4.0	5.0	18.0	20.0	19.0
5	17.0	16.0	16.5	36.0	36.0	36.0
10	38.0	35.0	36.5	61.0	62.0	61.5
20	49.0	48.0	48.5	68.0	70.0	69.0
30	49.0	50.0	49.5	70.0	72.0	71.0

1.4. Influence of different Photo-initiators and Polypropylene Mn on crosslinking

Table A5: Raw gel content data of UV cured PP (PP-67k) containing 2wt% DCP & 0.99 TR AOTEMPO with different photo-initiators at 0.5 wt% ([I2959] = 22.3 $\mu\text{mol/g}$, [I819] = 12 $\mu\text{mol/g}$, [L-TPO] = 14.4 $\mu\text{mol/g}$; [DCP] = 74 $\mu\text{mol/g}$, [AOTEMPO] = 146.4 $\mu\text{mol/g}$).

UV Time (mins)	I2959			I819			L-231		
	Trial 1	Trial 2	Average	Trial 1	Trial 2	Average	Trial 1	Trial 2	Average
1	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
3	18.0	20.0	19.0	7.0	5.0	6.0	6.0	8.0	7.0
5	36.0	36.0	36.0	29.0	27.0	28.0	27.0	28.0	27.5
10	61.0	62.0	61.5	58.0	55.0	56.5	54.0	56.0	55.0
20	68.0	70.0	69.0	64.0	63.0	63.5	63.0	62.0	62.5
30	70.0	72.0	71.0	68.0	69.0	68.5	68.0	68.0	68.0

Table A6: Raw gel content data of UV cured PP (PP-97k) containing 2wt% DCP & 0.99 TR AOTEMPO with different photo-initiators at 0.5 wt% ([I2959] = 22.3 $\mu\text{mol/g}$, [I819] = 12 $\mu\text{mol/g}$, [L-TPO] = 14.4 $\mu\text{mol/g}$; [DCP] = 74 $\mu\text{mol/g}$, [AOTEMPO] = 146.4 $\mu\text{mol/g}$).

UV Time (mins)	I2959			I819			L-231		
	Trial 1	Trial 2	Average	Trial 1	Trial 2	Average	Trial 1	Trial 2	Average
1	2.0	1.0	1.5	10.0	9.0	9.5	11.0	9.0	10.0
3	19.0	14.0	16.5	42.0	48.0	45.0	41.0	47.0	44.0
5	53.0	50.0	51.5	56.0	54.0	55.0	55.0	59.0	57.0
10	61.0	59.0	60.0	66.	67.0	66.5	63.0	62.0	62.5
20	68.0	66.0	67	69.0	70.0	69.5	66.0	65.0	65.5
30	72.0	69.0	70.5	71.0	70.0	70.5	71.0	70.0	70.5